Genome Intelligence

Michael Schatz

November 17, 2025
Applied Comparative Genomics

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Computational Genomics: Applied Comparative Genomics
Project Presentations

Presentations will be a total of 9 minutes: 7 minutes for the presentation, followed by 2 minutes for questions. We will strictly keep to the schedule to ensure that all groups can present in class!

Schedule of Presentations

Slot	Day	Start	Team	Students	Title
1	11/19	3:00	Spatial Dynamics	Dhanav Handa, Arijit Mahesh Kulkarni	Learning Cellular Perturbation Dynamics in Foundation-Model Embedding Space via Neural Optimal Transport
2	11/19	3:09	V for Vogelsteins	Suki Ogihara	Cost-Effective Cancer Diagnostic via Aneuploidy dection with RealSeqS
3	11/19	3:18	Iso-Scope	Sai Dharmasena	Iso-Scope: Benchmarking Gene Isoform Inference Methods for Single-Cell Data
4	11/19	3:27	MTeam	Malavika Nair	Assessing Data Completeness in Variant Pathogenicity Prediction
5	11/19	3:36	Team Causality	Ziwei Jiang, Zihan Zhou	Causal Representation Learning for Gene Expression
6	11/19	3:45	Kmer-Bin matrix	Zhuolun (Julian) Du	A Probabilistic Approach to Hi-C Sequence Localization
7	11/19	3:54	Team KMErs	Emily Sanchez Ocotlan, Maria Valverde-Cervantes, Kinverli Garcia Osorio	GenoViz: An Interactive Dashboard for Comparative Visualization of Personal Genomic Data
1	12/1	3:00	Team Zadorozhny	Nikola Zadorozhny	Fine-Tuning DNA Foundation Models for Variant Effect Prediction
2	12/1	3:09	AML Detection LLM	Charlotte Cheung, Claire Cui	Fine-tuning a Biomedical Language Model for Leukemia Subtype Prediction from Gene Expression Data
3	12/1	3:18	PrecisionMut	Rongrong Yan	Benchmarking Non-coding Mutation Impact Prediction in Colorectal Cancers Using Multi-Method Functional Scores
4	12/1	3:27	Team RNN	Alex Dong, Melody Jin, Katie Hong	Identification of Bacterial Promotors
5	12/1	3:36	Yolks to Folks	Megan Miller	Searching for Genetic Signatures and Constraints of Viviparity
6	12/1	3:45	Team Chauhan	Akshat Chauhan	Residual Variational Autoencoder for De Novo Design of Tumor-Specific Promoters
7	12/1	3:54	Julie X and Jason Z	Julie Xian, Jason Zavras	Evolutionary origin of Human Disease Genes: A Comparative Genomics Approach
8	12/1	4:03	LettUCE Grow	Xinyi Chen	Age vs Stress-induced inflammation: Universal Cell Embedding (UCE)-Based Comparative Profiling
1	12/3	3:00	Team Dark Magic	Ezra Greenberg	Analysis of MAGE Dataset for Microbial Contamination
2	12/3	3:09	Multiomics	Matthew Wong	Integrated analysis of Multiomics dataset for colon cancer patients; unravelling key prognostic biomarkers
3	12/3	3:18	MonoMarker	Fuchen Li, Huanchang Tu	Benchmarking single-cell trajectory inference under varying sequencing coverage using Monocle3
4	12/3	3:27	MetaGen	Charissa Luk, Rachael Pei, Jessie Huh	Predicting Type 2 Diabetes from Gut Microbiome Composition Through Classifier Benchmarking and Machine Learning
5	12/3	3:36	The Pink Helix	Lauren Ji, Sofia Floody	Investigating the Rate of Heterozygosity and Breast Cancer Among Global Populations
6	12/3	3:45	STalign2	Iris Kwon, Smriti Grounder	Improving STalign Accuracy by Incorporating Gene Expression
7	12/3	3:54	Chroma-Chrunchers	Martin Hilser	How Little Detail is Enough? Evaluating Chromatin-State Annotations Robustness Under Low-Coverage Sequencing
8	12/3	4:03	Relish	Reena Assassa, Lillianna Gund, Shaun Ku	Comparing Genomic LLM Variant Pathogenicity Predictions with Established In-Silico Scores

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https://github.com/schatzlab/appliedgenomics2025/blob/main/project/presentations.md

Recommended outline for your talk (30 seconds to 1 minute per slide):

1. Title Slide: Who are you, title, date
2. Intro: Whats the big idea???
3. Data: What data are you looking at?
4. Methods 1: What did you try?
5. Methods 2: What is the key idea?
6. Results 1: What did you see!
7. Results 2: How does it compare to other methods/data/ideas?
8. Discussion: What did you learn from this study? What are the next steps
9. Acknowledgements: Who helped you along the way?
10. Thank you!

I strongly *discourage* you from trying to give a live demo as they are too unpredictable for a short talk. If you have new software you want to show, use a "cooking show" approach, where you have screen shots of the important steps.

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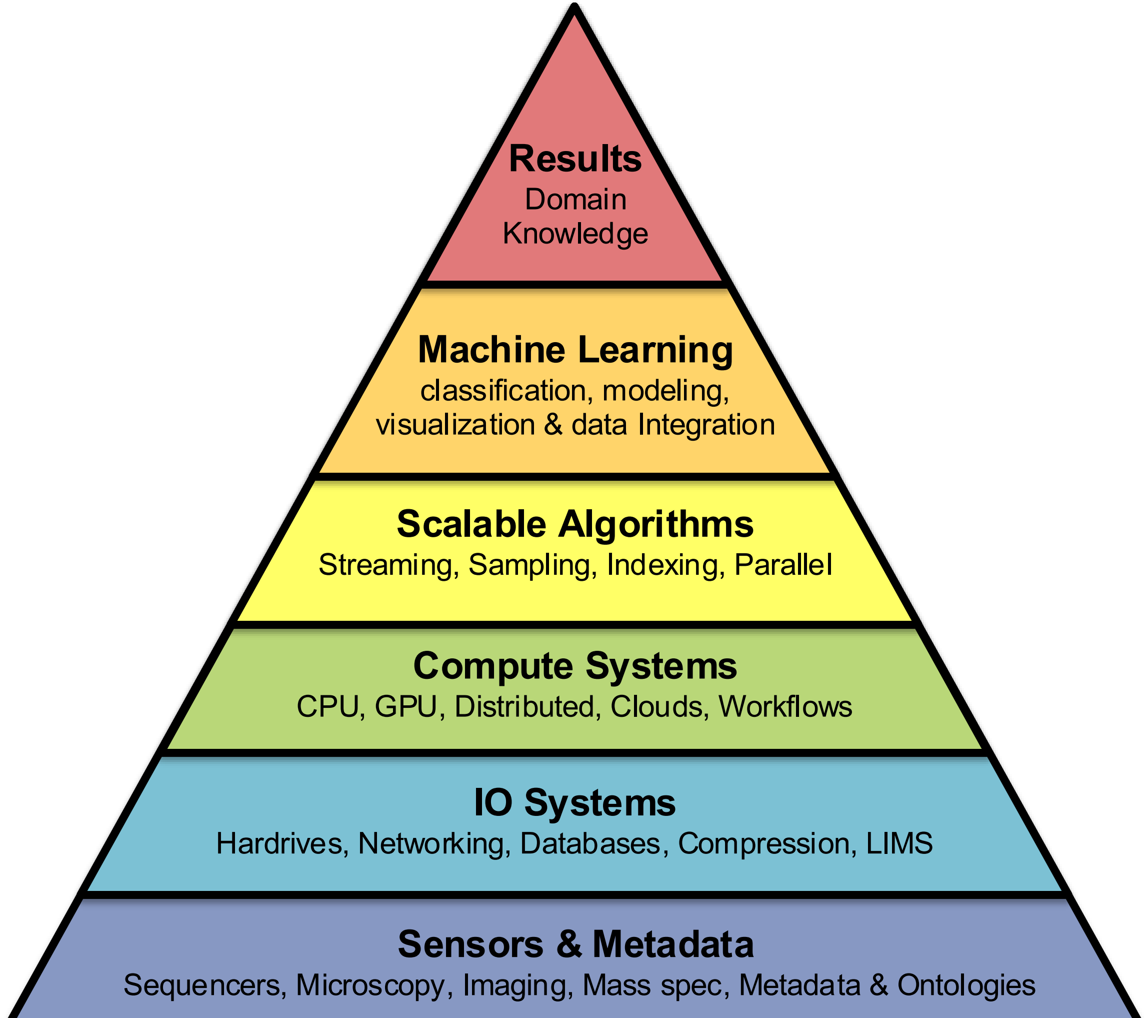
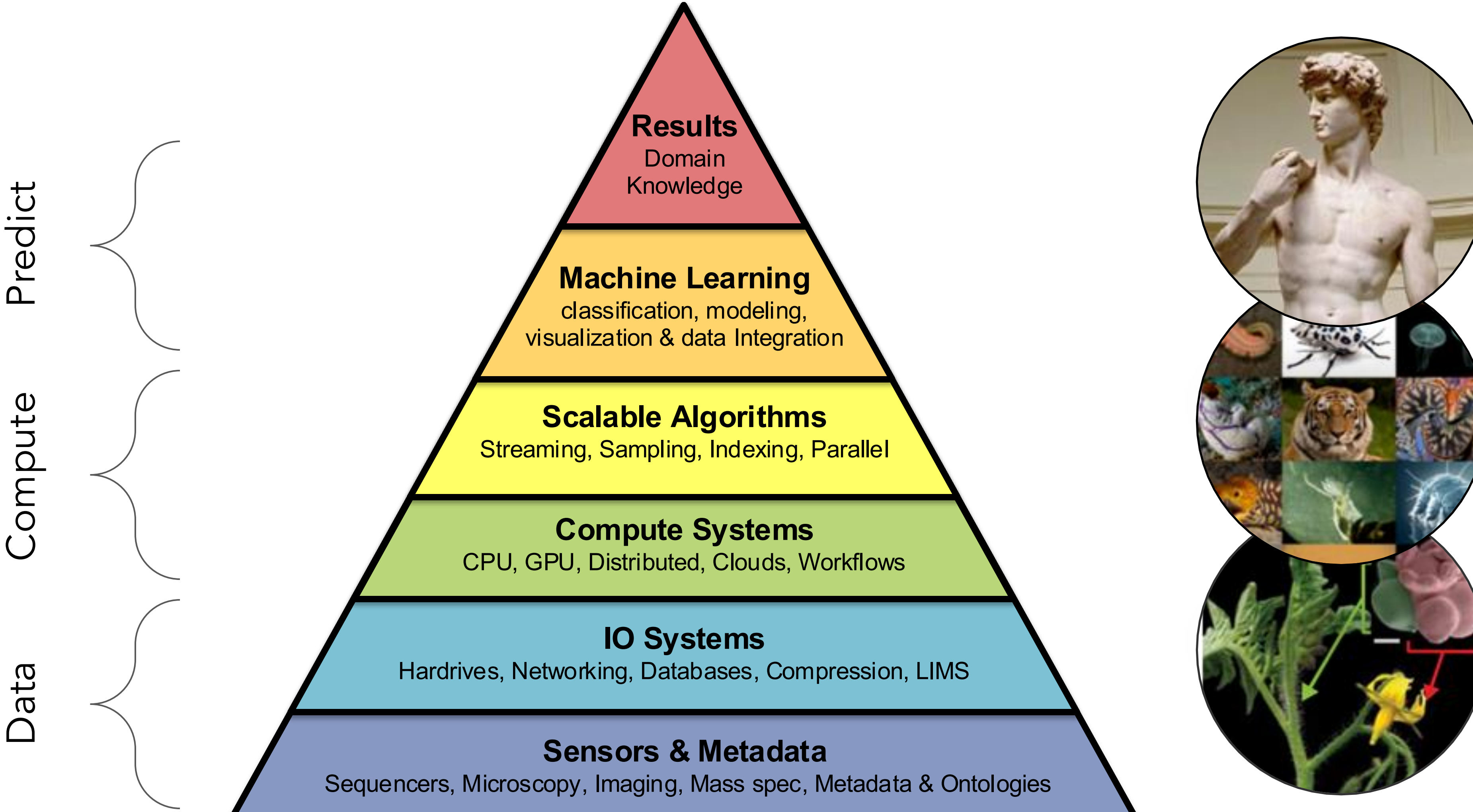
DALL-E: make an image of michael schatz pondering the relationship between genotype and phenotype



Biological data sciences in genome research

Schatz MC (2015) Genome Research doi: 10.1101/gr.191684.115



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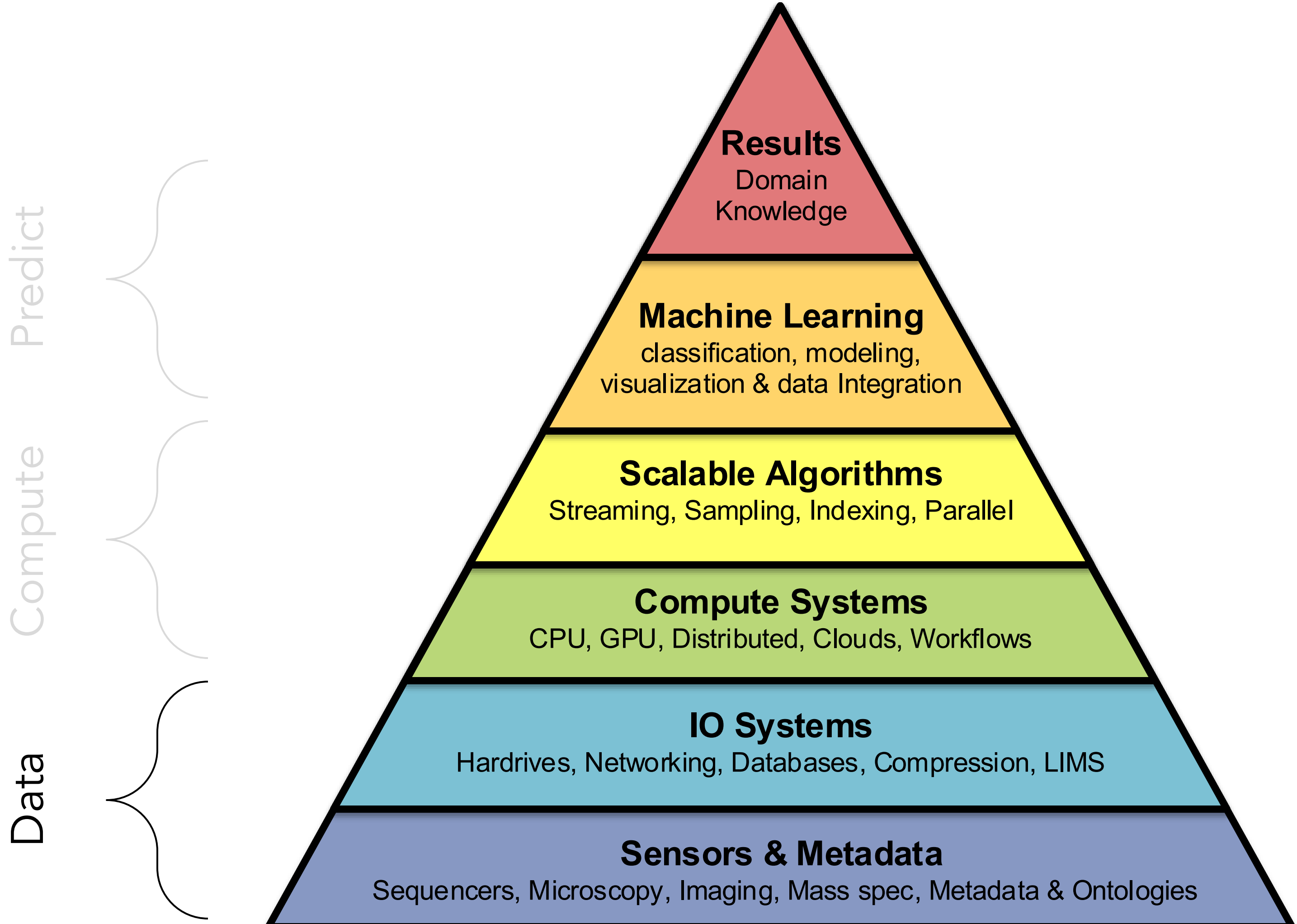

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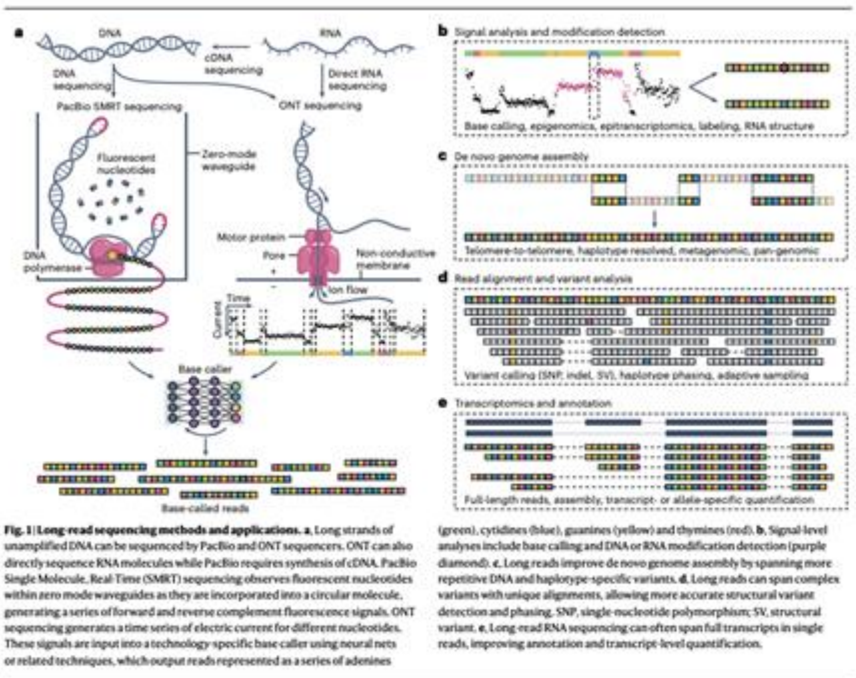
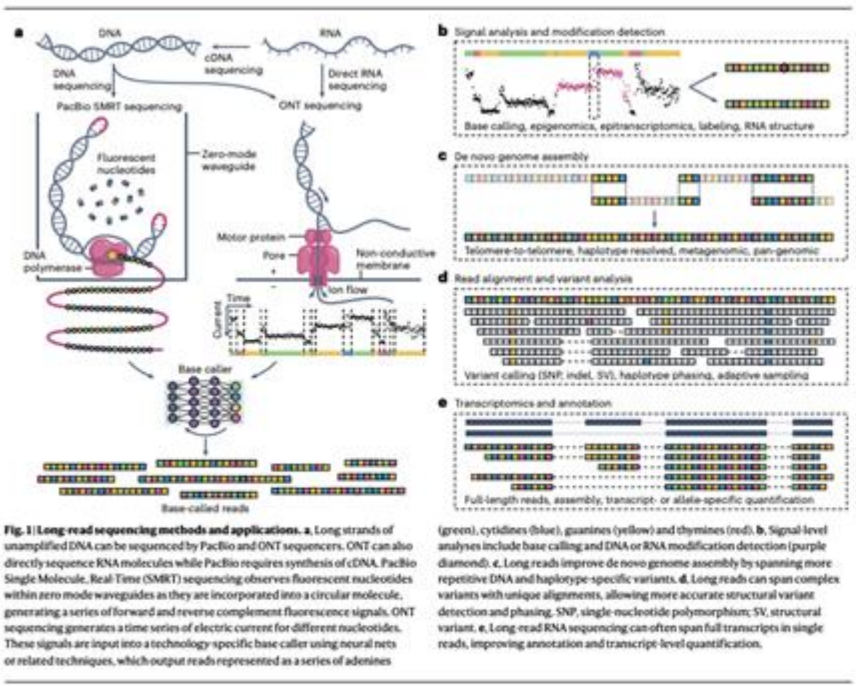


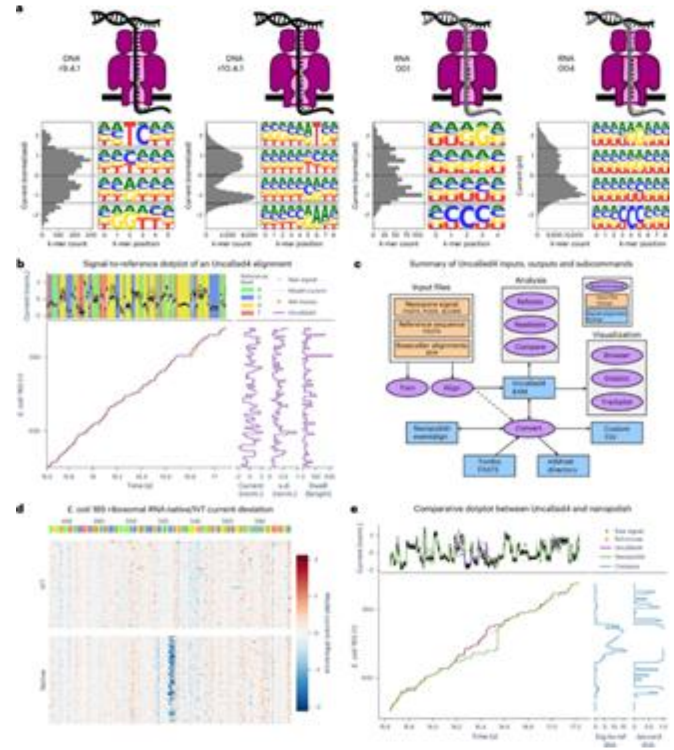
Fig. 1 | Long-read sequencing methods and applications. a, Long strands of unamplified DNA can be sequenced by PacBio and ONT sequencers. ONT can also directly sequence RNA molecules while PacBio requires synthesis of cDNA. PacBio Single Molecule, Real-Time (SMRT) sequencing observes fluorescent nucleotides within zero mode waveguides as they are incorporated into a circular molecule, generating a series of forward and reverse complement fluorescence signals. ONT sequencing generates a time series of electric current for different nucleotides. These signals are input into a technology-specific base caller using neural nets or related techniques, which output reads represented as a series of adenines (green), cytidines (blue), guanines (yellow) and thymines (red). **b**, Signal-level analyses include base calling and DNA or RNA modification detection (purple diamond). **c**, Long reads improve de novo genome assembly by spanning more repetitive DNA and haplotype-specific variants. **d**, Long reads can span complex variants with unique alignments, allowing more accurate structural variant detection and phasing. SNP, single-nucleotide polymorphism; SV, structural variant. **e**, Long-read RNA sequencing can often span full transcripts in single reads, improving annotation and transcript-level quantification.

Approaching complete genomes, transcriptomes and epi-omes with accurate long-read sequencing

Kovaka *et al* (2023) *Nature Methods*.

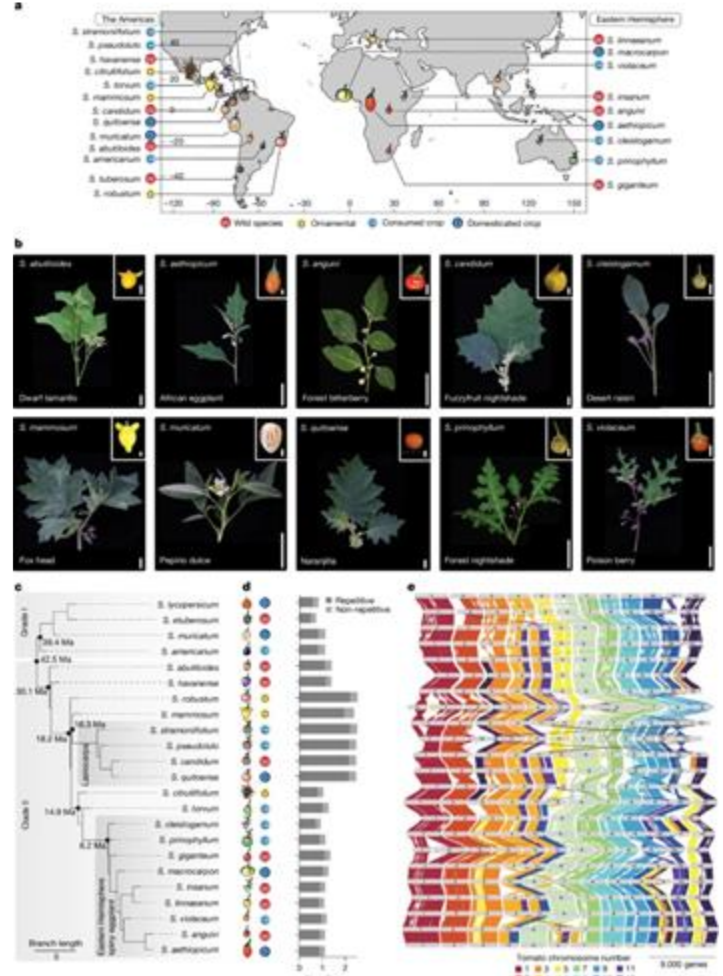
https://doi.org/10.1038/s41592-022-01716-8





Approaching complete genomes, transcriptomes and epi-omes with accurate long-read sequencing
Kovaka *et al* (2023) *Nature Methods*.
https://doi.org/10.1038/s41592-022-01716-8

Uncalled4 improves nanopore DNA and RNA modification detection via fast and accurate signal alignment
Kovaka *et al* (2025) *Nature Methods*.
https://doi.org/10.1038/s41592-025-02631-4



Solanum pan-genetics reveals paralogues as contingencies in crop engineering
Benoit, Jenike, et al (2025) Nature. https://doi.org/10.1038/s41586-025-08619-6

Pan-genomics and pan-genetics enables the reciprocal exchange of genotype-to-phenotype knowledge between major crops and indigenous crops

Tomato

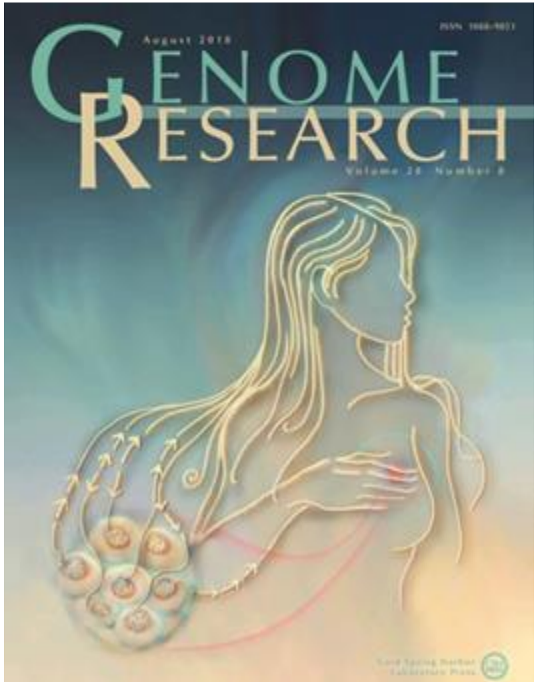
CLV3

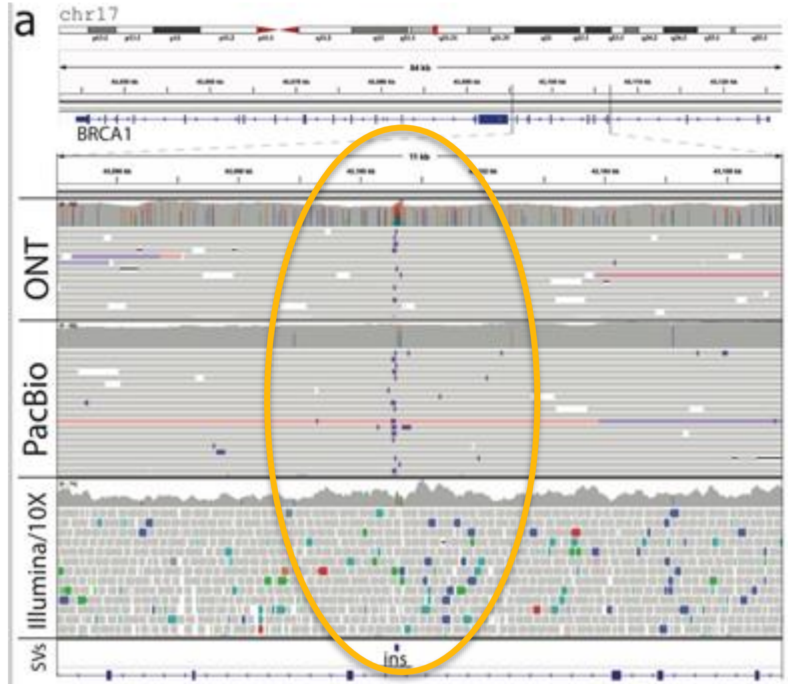
SCPL25

African eggplant

- **Paralog dynamics are major drivers of trait variation**
- **Translating knowledge between species (pan-genetics) is contigent on understanding paralogs**
- **Major opportunities for enhancing the diversity and resilience of our food system**

Hidden Variants in Breast Cancer Genes





62bp repeat expansion in BRCA1 in germline tissue that is undetectable using short read sequencing

Complex rearrangements and oncogene amplifications revealed by long-read DNA and RNA sequencing of a breast cancer cell line
Nattestad *et al* (2018) Genome Research doi: 10.1101/gr.231100.117

Comprehensive analysis of structural variants in breast cancer genomes using single molecule sequencing
Aganezov *et al.* (2020) Genome Research. doi:10.1101/gr.260497.119

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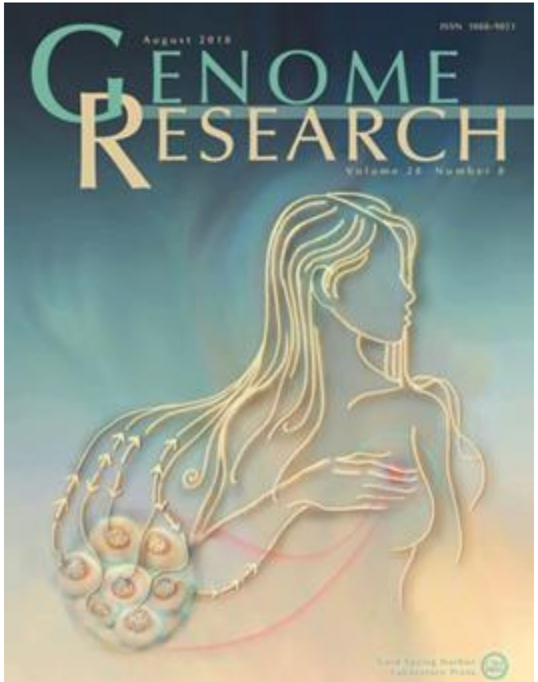

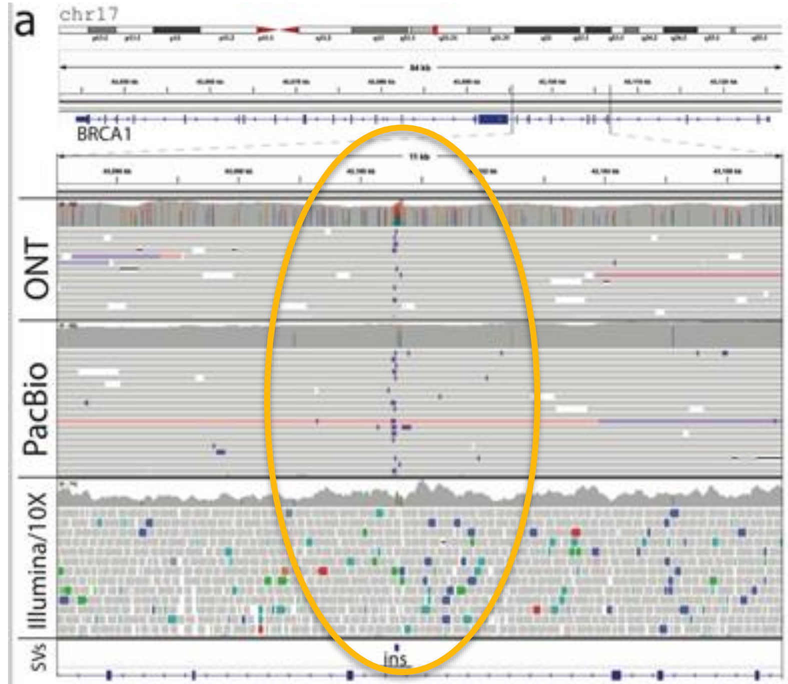

How can we identify variations with clinical & functional impact?

Complex rearrangements and oncogene amplifications revealed by long-read DNA and RNA sequencing of a breast cancer cell line
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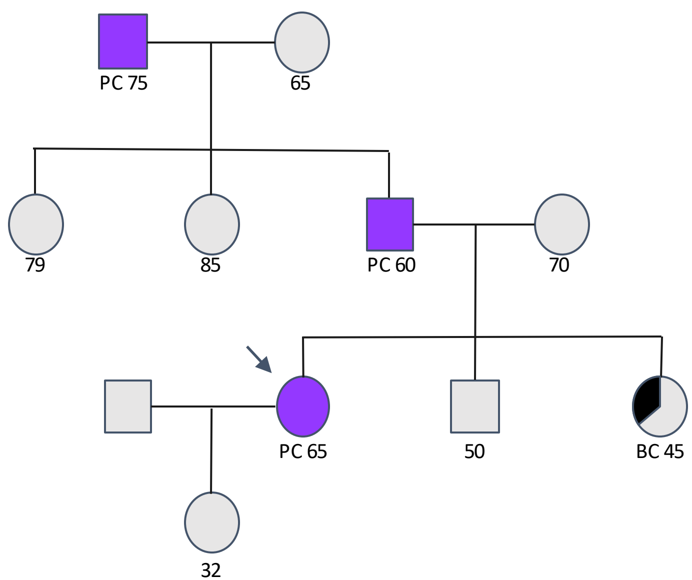
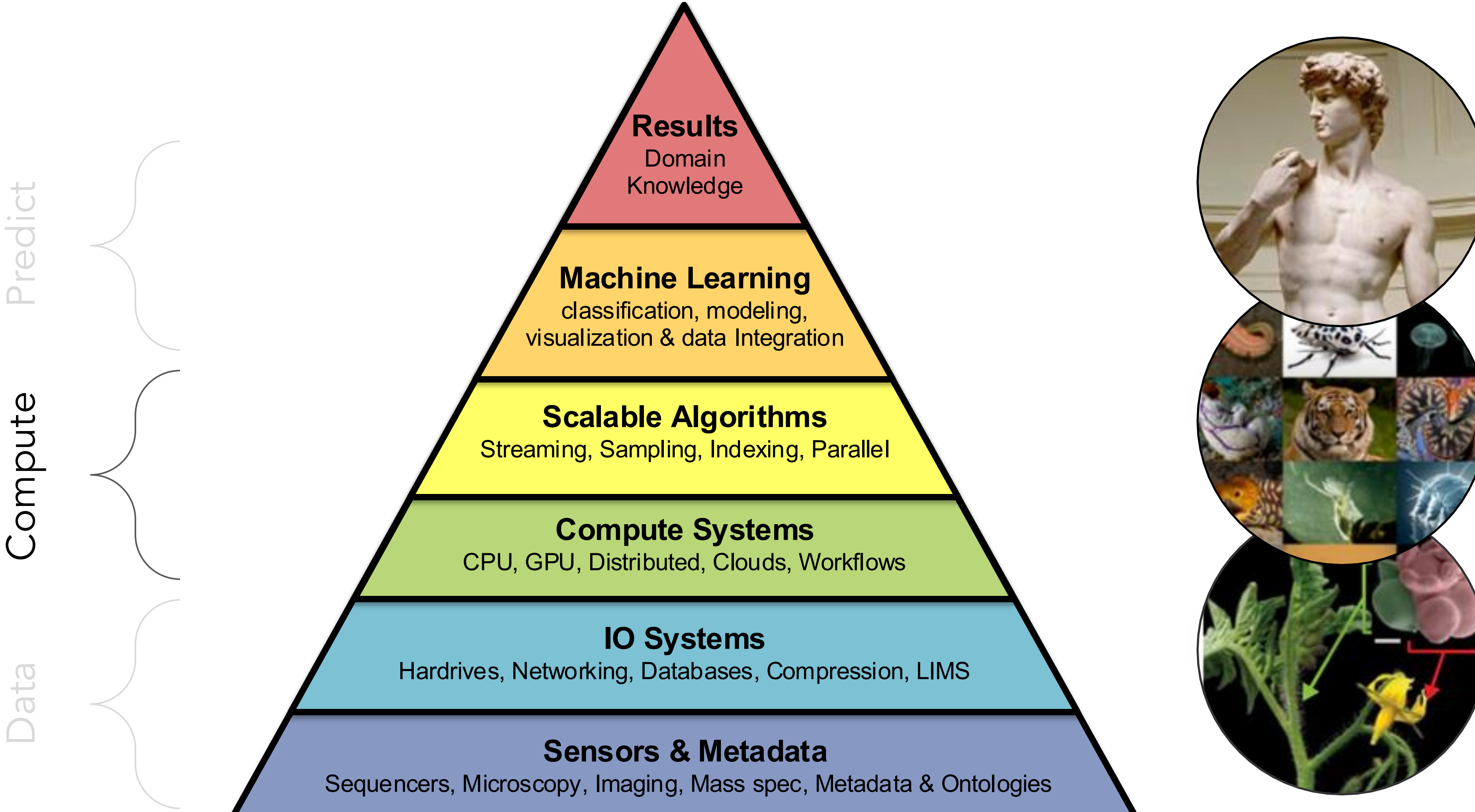
Study Design



* Representative Pedigree

- Families enrolled from ongoing registries of Pancreatic Cancer Families at:
  - Johns Hopkins: Alison Klein (+Winston Timp)
  - University of Toronto: Steven Gallinger
  - Mayo Clinic: Gloria Petersen/Sam Antwi
- Selection for families with multiple pancreatic cancers (3+) and or 2 pancreatic cancer and strong history of other cancers
- Exclude families with known pathogenic variants in known pancreatic cancer risk genes:
  - BRCA2, PALB2, BRCA1, ATM, CDKN2A, PRSS1, STK11, MSH2, MLH1T, P53, PMS2, MSH6



Biological data sciences in genome research

Schatz MC (2015) Genome Research doi: 10.1101/gr.191684.115

The Galaxy Platform for Accessible, Reproducible and Collaborative Biomedical Analyses

http://usegalaxy.org

Total registered users
637K

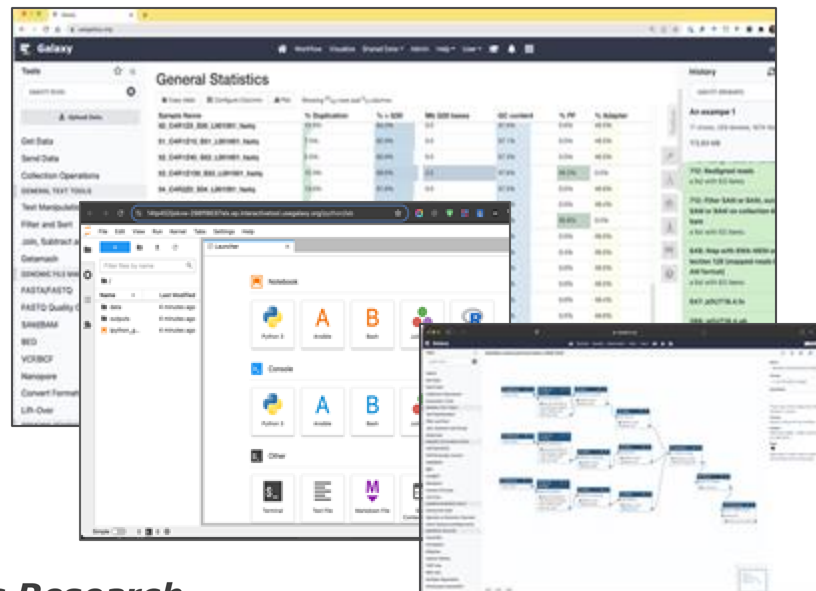
Total Jobs
183M

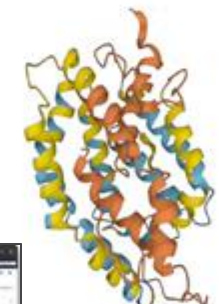
Total datasets
327M

Total Workflows
846K

Tutorials
424

Citations
22,428

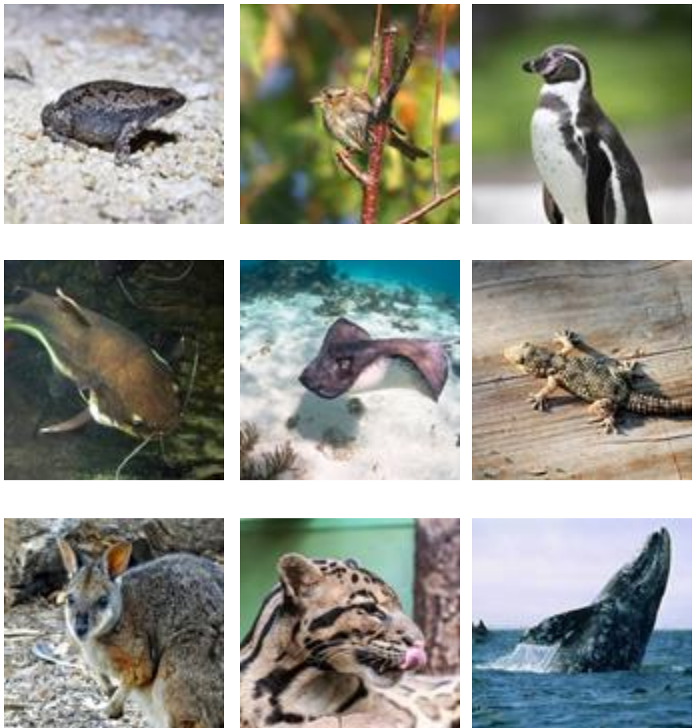


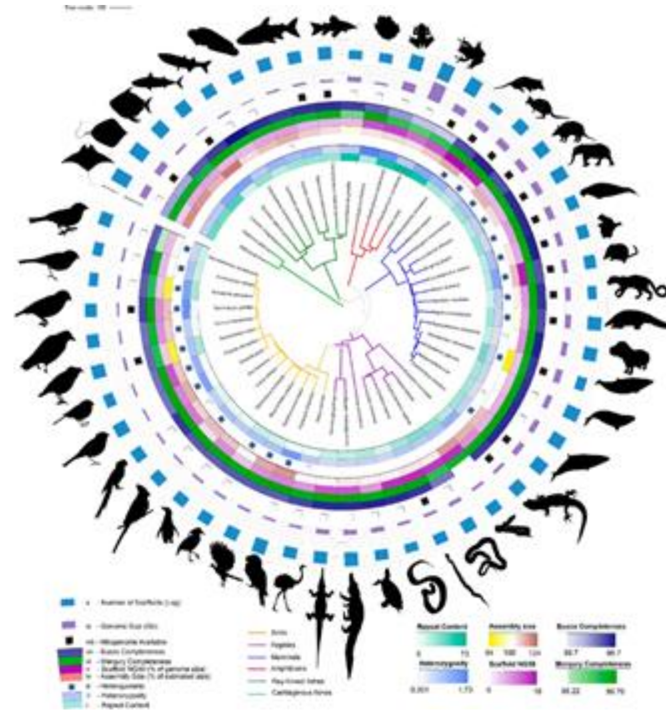


The Galaxy Community (2024) ***Nucleic Acids Research***
https://doi.org/10.1093/nar/gkae410

VGP VERTEBRATE GENOMES PROJECT

A PROJECT OF THE G10K CONSORTIUM



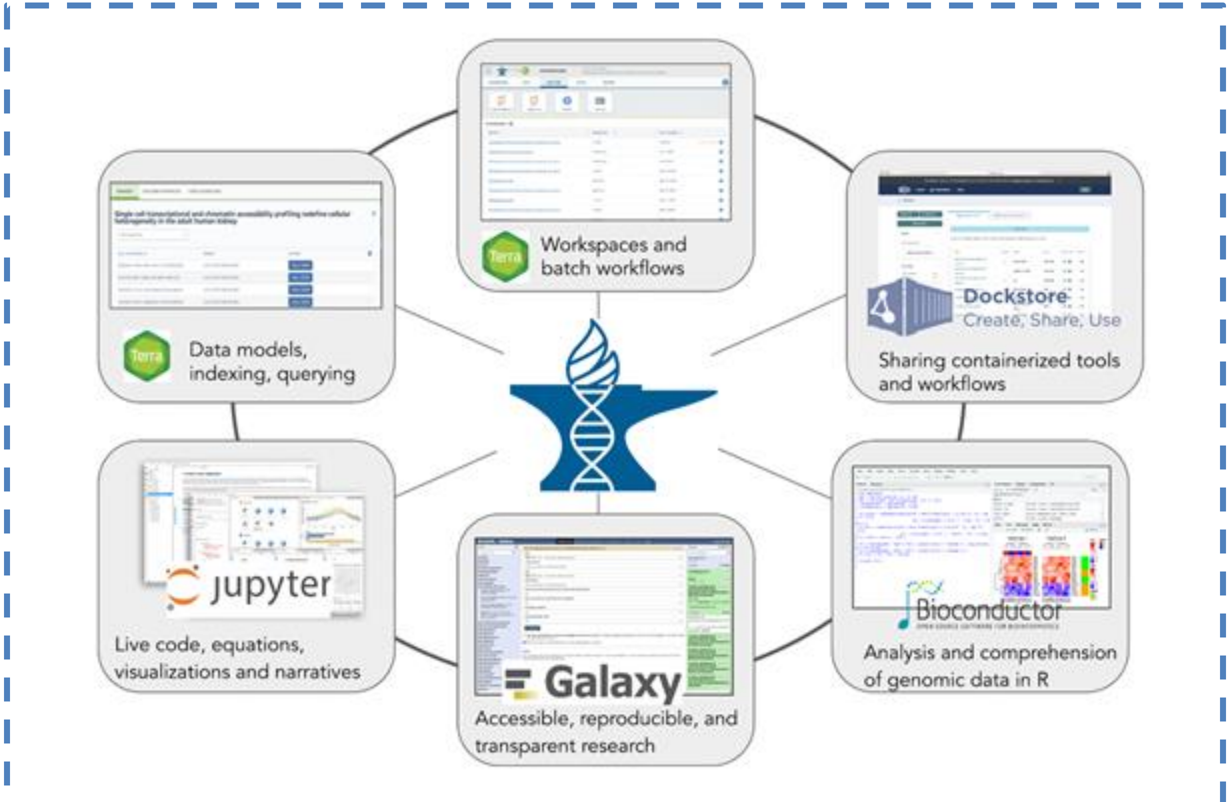


Scalable, accessible, and reproducible reference genome assembly and evaluation in Galaxy
Lariviere *et al.* (2024) Nature Biotechnology.
https://doi.org/10.1038/s41587-023-02100-3

Phase I complete! 217 orders (80% of all vertebrate orders)

Galaxy PROJECT

AnVIL: A Secure Federated Data & Compute Ecosystem



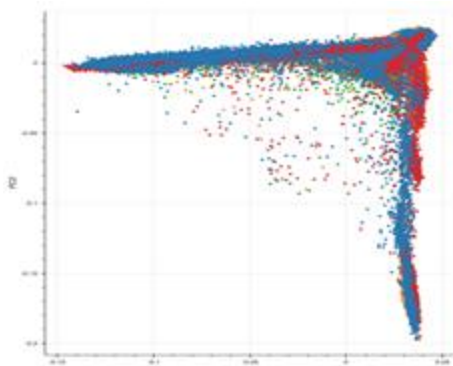
NIH
NHGRI

FedRAMP

Over 5 petabytes, 600,000 genomes and growing every day!

Powered by AnVIL

Centers for Common Disease Genomics



<u>136,959 WGS samples</u>

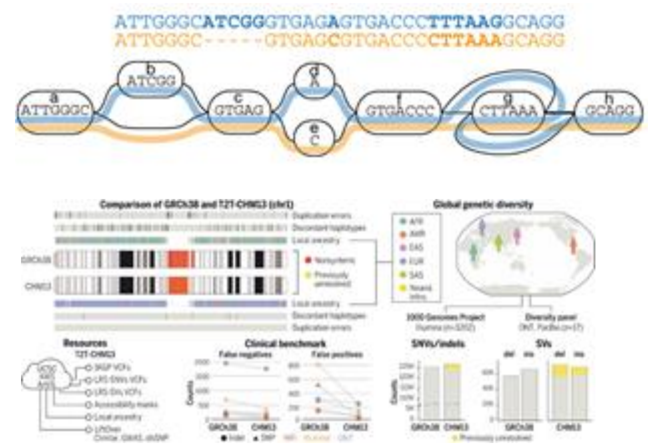
Baylor College: 35,493

Broad Institute: 16,828

NYGC: 40,975

WashU: 43,620

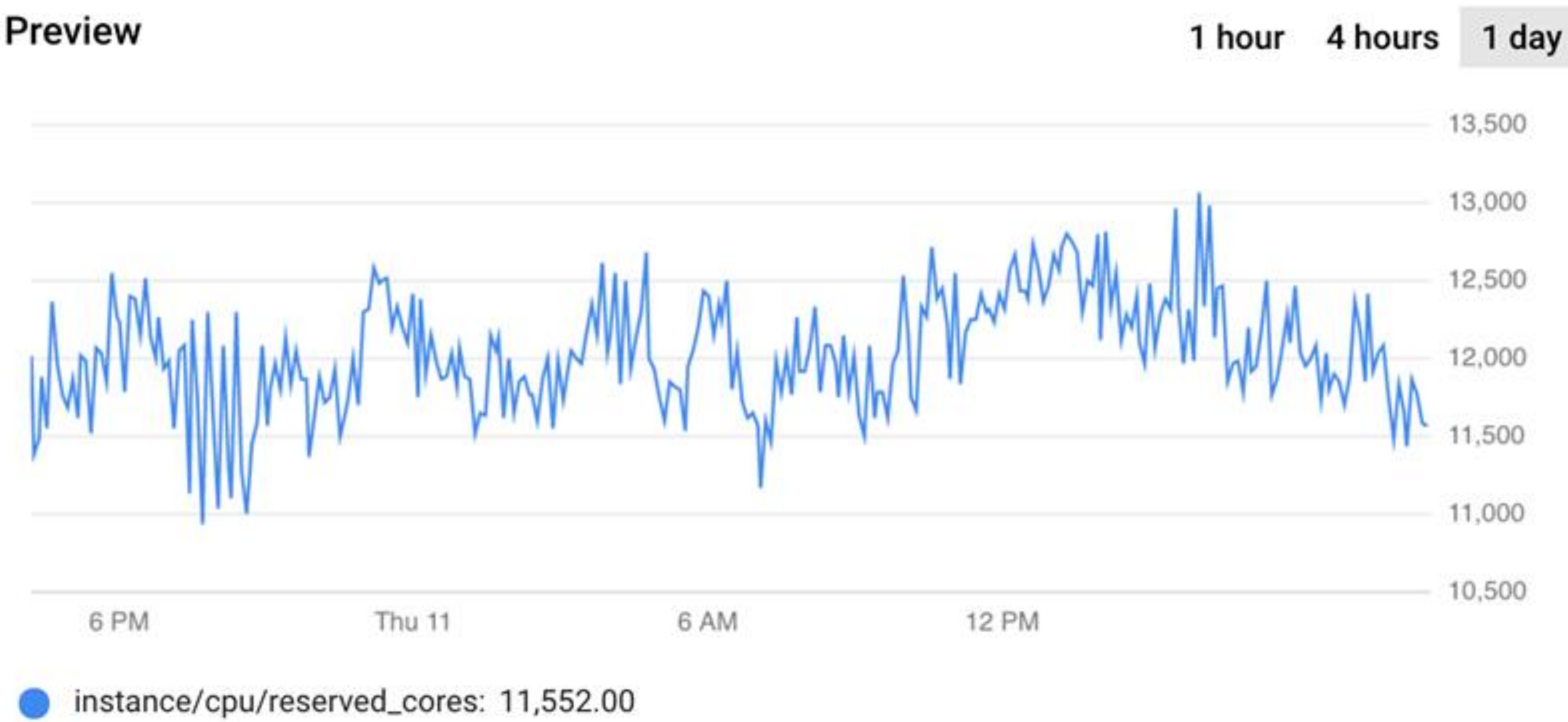
TOWARDS A COMPLETE REFERENCE OF HUMAN GENOME DIVERSITY



Nurk et al. (2022) Science

Liao et al. (2023) Nature

Core usage over 24 hours




Samantha Zarate

https://dockstore.org/workflows/github.com/schatzlab/t2t-variants/T2T_alignment

A complete reference genome improves analysis of human genetic variation
Aganezov, S*, Yan, SM*, Soto, DC*, Kirsche, M*, Zarate, S*, *et al.* (2022) *Science.* doi: 10.1126/science.abl3533

All of Us Research Program: Advancing Precision Medicine

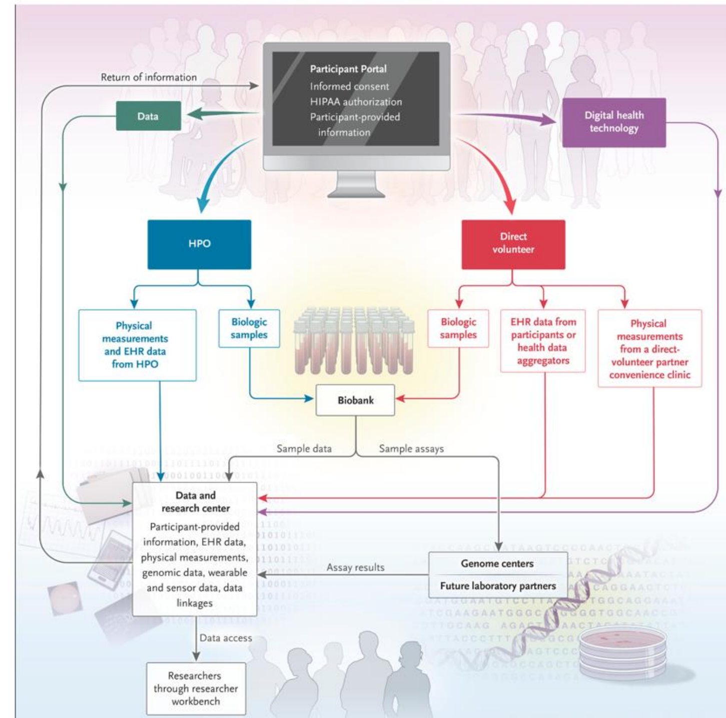
Why start *All of Us*?

- Genomic, Electronic Health Records (EHR), and other personalized data for 1M US citizens
- One-size-fits-all care often falls short for individual patients
- Fragmented health records limit coordinated, holistic care
- Redundant data collection slows discovery and increases cost

What we aim to achieve?

- Identify risk factors for both common and complex diseases
- Understand which treatments work best for people of different backgrounds
- Match participants to clinical studies that fit their needs
- Explore how technology can support personalized health

From promise to progress --
Shaping the future of health research



The "All of Us" research program
All of Us Research Program Investigators (2019) *NEJM*
doi: 10.1056/NEJMsr1809937

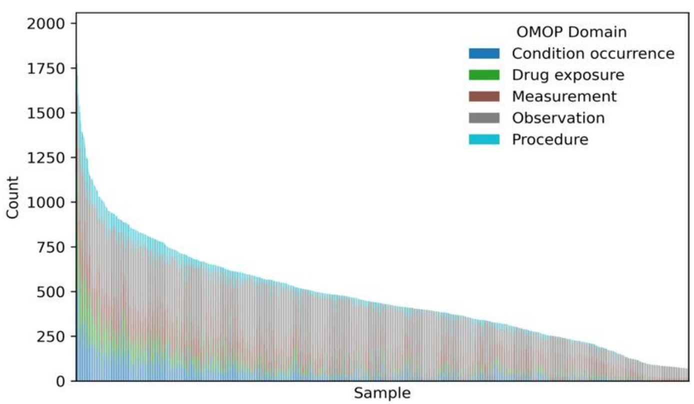
All of Us Long Read Sequencing Participants

Sample information

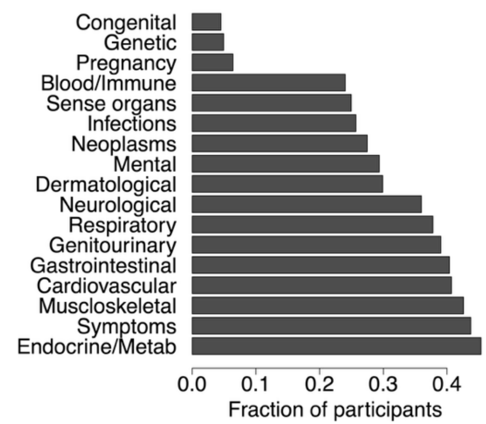
- 1,027 self-identified Black or African American participants
- Selected from the All of Us Research Program (CDR v7), which includes 245,388 short-read participants across the United States

Electronic health record data

- Clinical data mapped to five domains of the OMOP Common Data Model (CDM)



- Disease status among the 1,027 AoU long-read participants



Variant Discovery

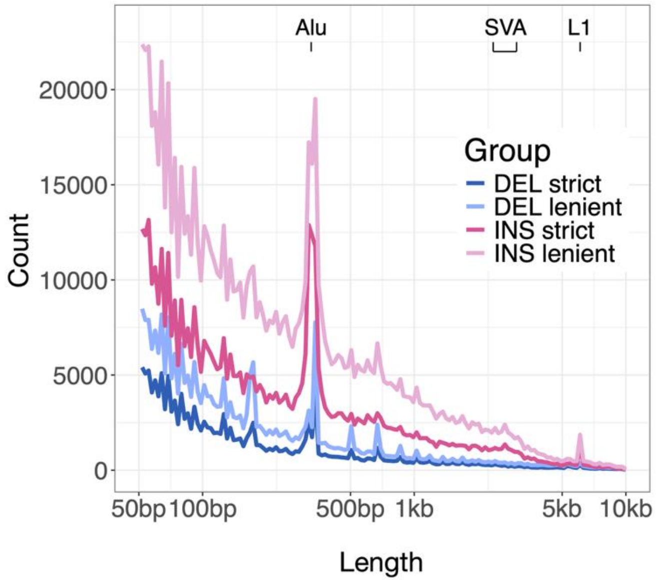
Samuel K Lee

Steve Huang

Fabio Cunial

Sample: 1027 long-read *AoU* samples + 47 HPRC samples, sequenced at 8× depth.

Distribution of SV lengths for insertions and deletions



Tools

- PAV, Sniffles2, and PBSV

Process

- Applied Truvari for per-sample clustering and normalization.
- Trained a per-sample XGBoost classifier trained to remove false positives with Kanpig
- Generated cohort-wide SV callsets by merging filtered per-sample callsets using Truvari

- **Lenient callset**: 1,213,876 SVs - prioritized for sensitivity
- **Strict callset**: 665,869 SVs - prioritized for high specificity

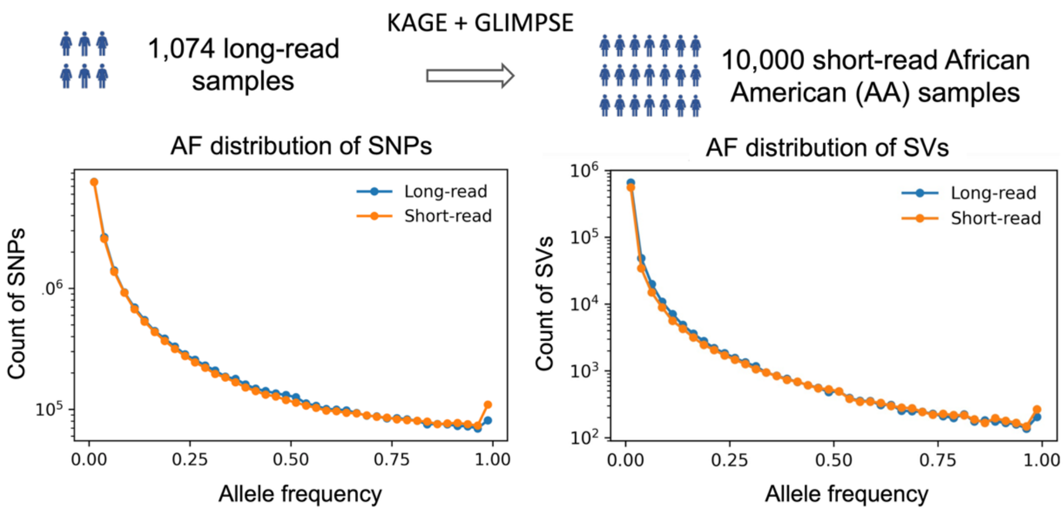
Phenome-wide SV Analysis Identifies Disease Risk Factors

Qiuhui Li

Objective

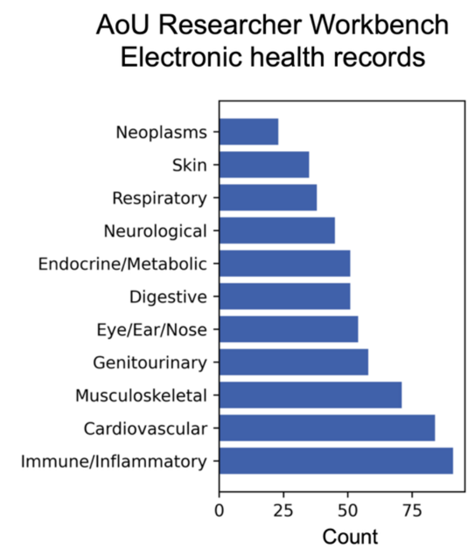
- Evaluate the clinical impact of SVs by integrating genomic data with electronic health records from an expanded cohort.

Genotyping and imputation with the long-read panel



18.8M SNPs and 773K SVs identified in 1,074 long-read samples
18.5M SNPs and 650K SVs genotyped in 10K short-read AA samples

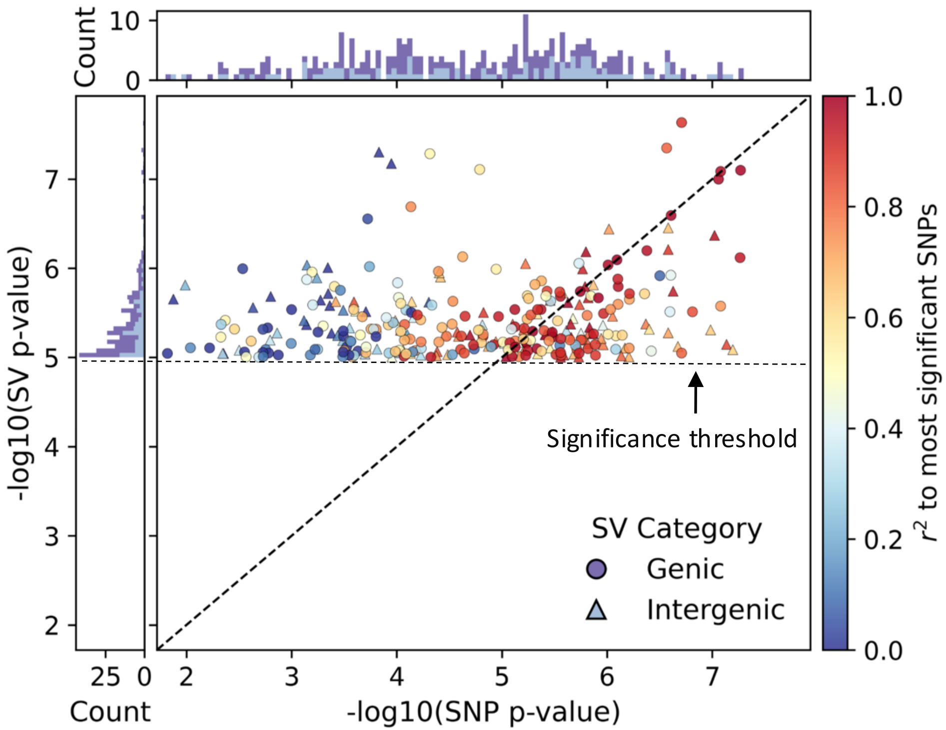
Phenotypes extraction & classification



492 disease phenotypes

Genome-Wide SV-Disease Associations

Comparison of the association significance between each SV and the strongest nearby SNP (within ±100 kb) for the same phenotype



- 291 SV–disease associations across 226 diseases.
- 148 associations (51%) involved SVs absent from the matched short-read callset derived from 990/1,027 of the same participants.
- 191 (66%) SV-disease pairs across 160 traits where the SV was the most likely causal variant within the locus.

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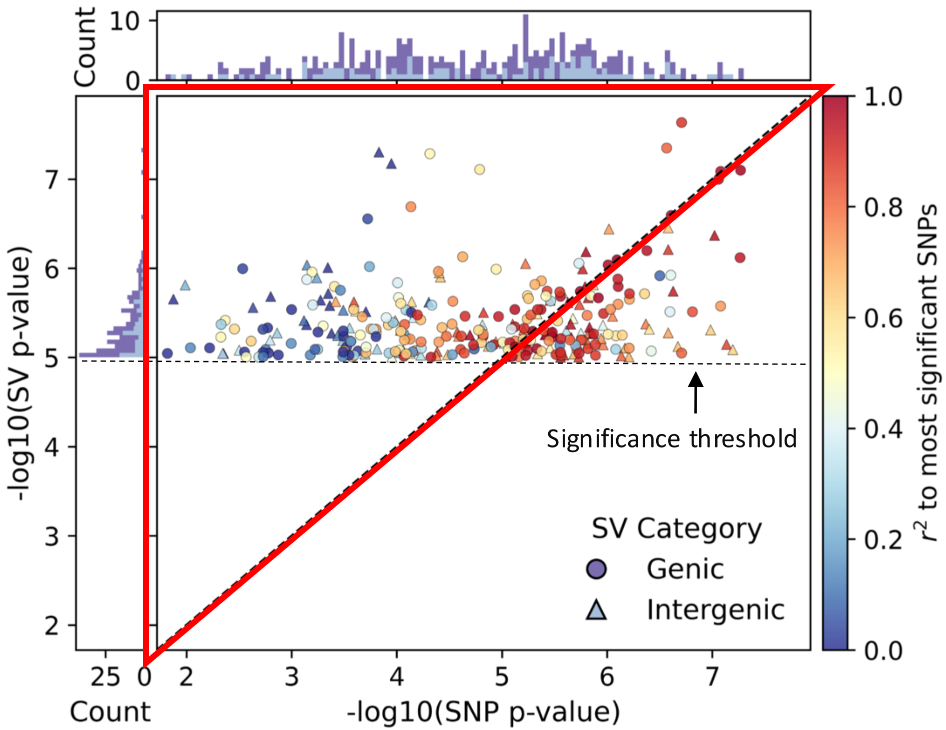

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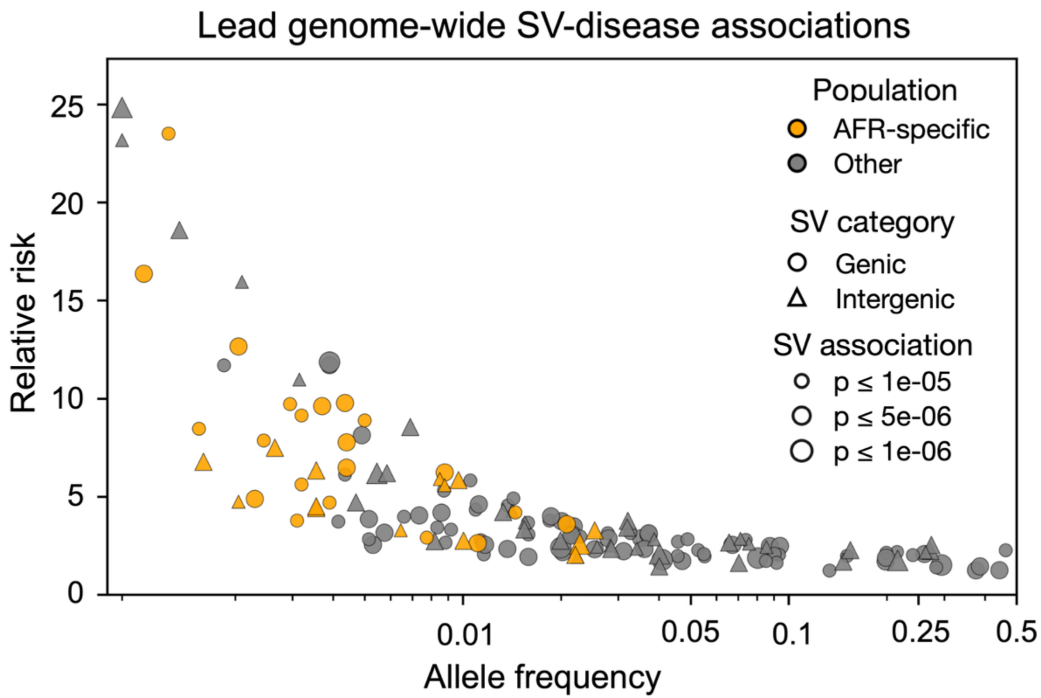
Genetic Architecture of Disease Risk

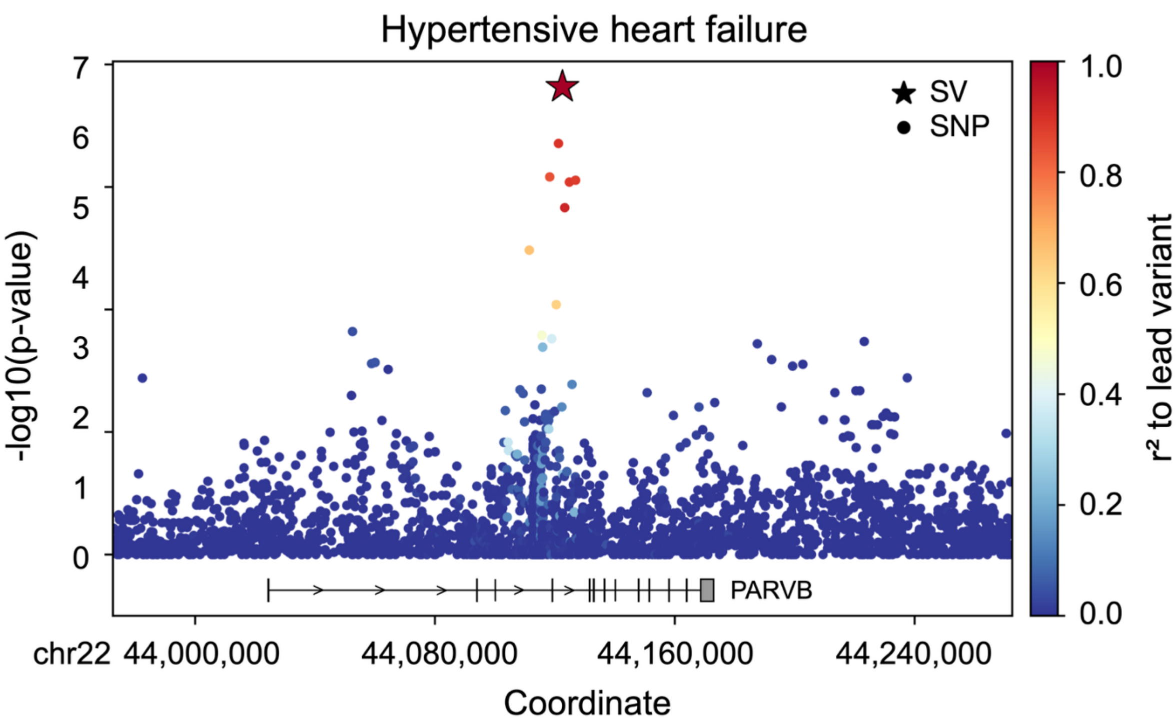
Relationship between allele frequency and relative risk (odds ratio > 1)

- Lead genome-wide SV-disease associations where the SV showed stronger signals than nearby SNPs.

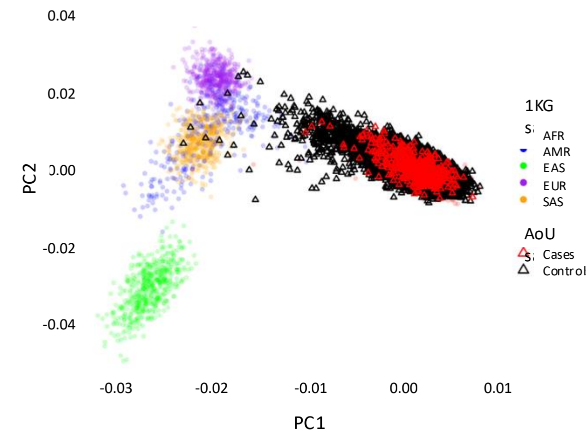


- Most high-effect associations occurred at low allele frequencies, suggesting potential purifying selection.

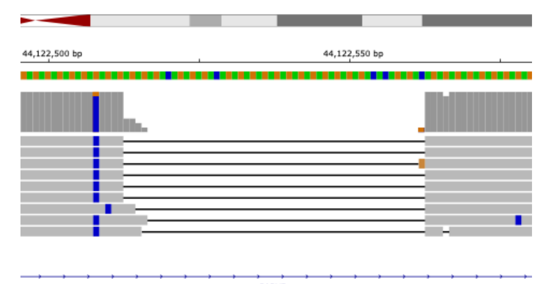
- A 50bp deletion in *PARVB* linked to hypertensive heart failure (OR = 1.87; 95% CI: 1.52–2.31; $p = 2.32\times10^{-8}$)
- *PARVB* encodes β-parvin, a protein that regulates cardiomyocyte shape and sarcomere assembly [1]



Samples with hypertensive heart failure: 511



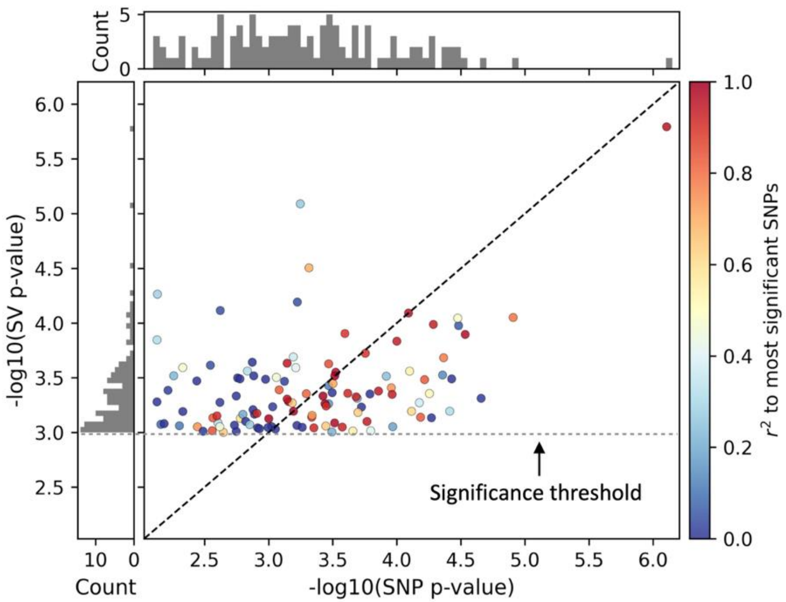
IGV screenshot of the deletion



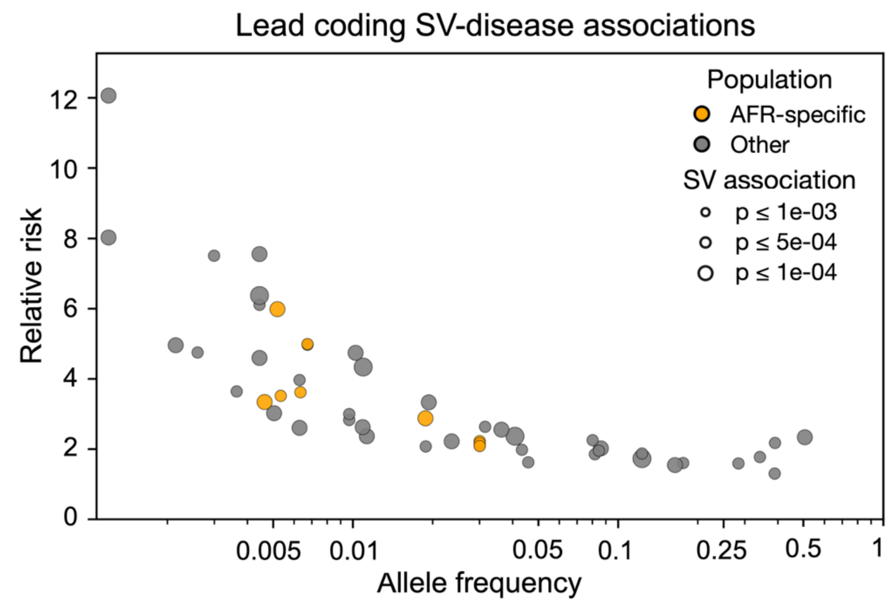
[1] Thievessen, I. et al. The focal adhesion protein β-parvin controls cardiomyocyte shape and sarcomere assembly in response to mechanical load. Curr. Biol. 32, 3033–3047.e9 (2022)

Disease Associations with SVs in Coding Regions of Genes

Comparison of SV association strength with strongest nearby SNV (within ±100 kb)



Relationship between allele frequency and relative risk



- 114 associations ($p < 1\times10^{-3}$) , 67 were linked to SVs absent from the short-read callset.
- In 62 of the 114 associations (54%), SVs showed stronger association signals than nearby SNVs within 100 kbp, including 40 involving SVs not detected in short-read data

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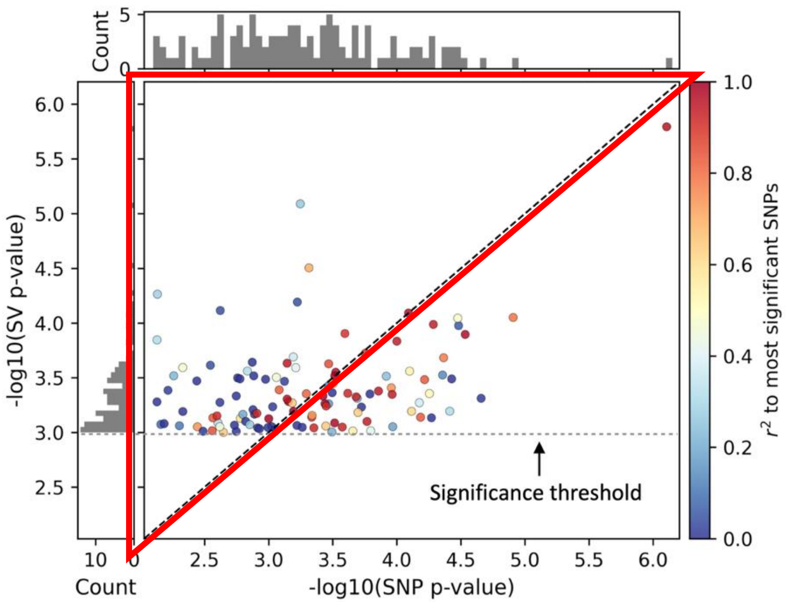
Comparison of SV association strength with strongest nearby SNV (within ±100 kb)



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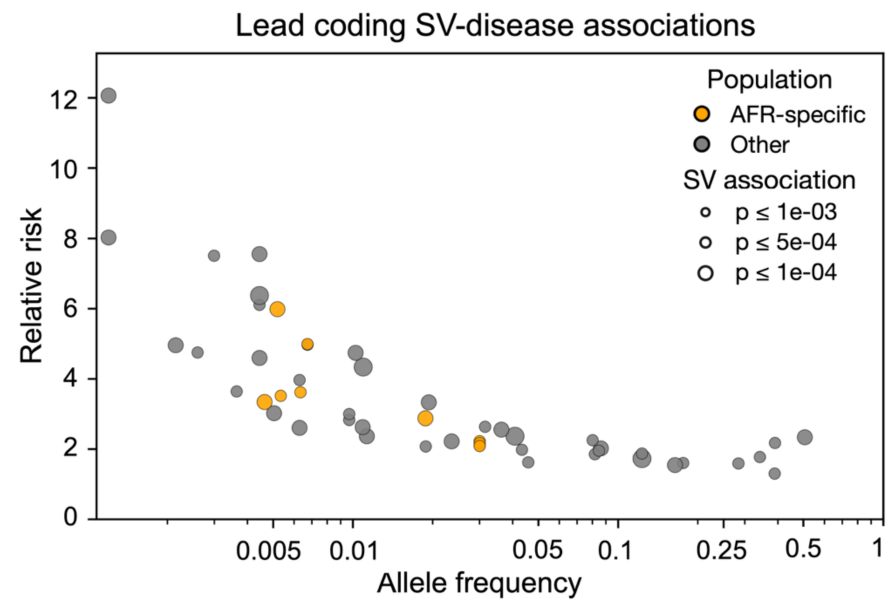
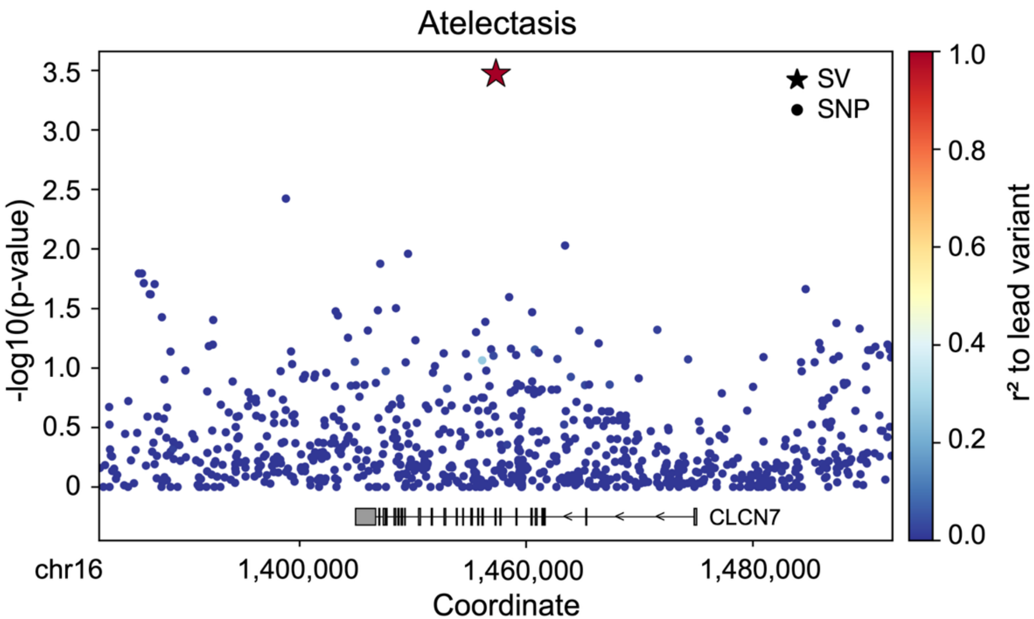
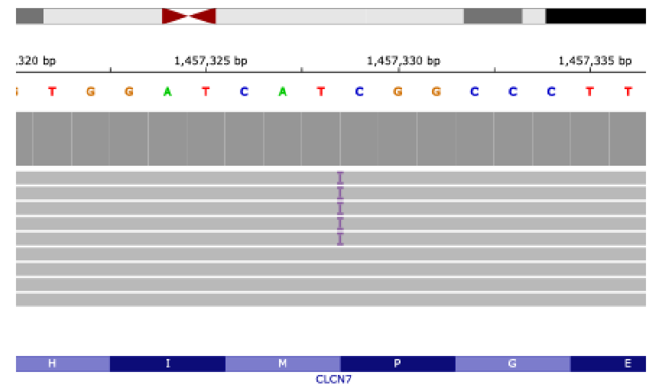


- 114 associations ($p < 1 \times 10^{-3}$) , 67 were linked to SVs absent from the short-read callset.
- In 62 of the 114 associations (54%), SVs showed stronger association signals than nearby SNVs within 100 kbp, including 40 involving SVs not detected in short-read data

Protein-level impact of SV in *CLCN7*

- A 200 bp insertion in *CLCN7* associated with atelectasis (Collapsed Lung) (OR = 2.60; 95% CI: 1.62-4.18; $p = 3.3\times10^{-4}$)





- The SV introduced a premature stop codon, resulting in a truncated protein lacking key functional domains
- *CLCN7* mutations have been reported to cause severe alveolar atelectasis in mice [1], providing mechanistic support for this finding in humans

[1] Maurizi, A. et al. Extra-skeletal manifestations in mice affected by Clcn7-dependent autosomal dominant osteopetrosis type 2 clinical and therapeutic implications. Bone Res. 7, 17 (2019)

Summary

Comprehensive SV landscape in African Americans

- 68.2% of deletions and 63.4% of insertions missed by short-read WGS

Prioritization of pathogenic variants

- 473 potentially pathogenic variants identified
- 273 absent from previous long-read callsets

PheWAS

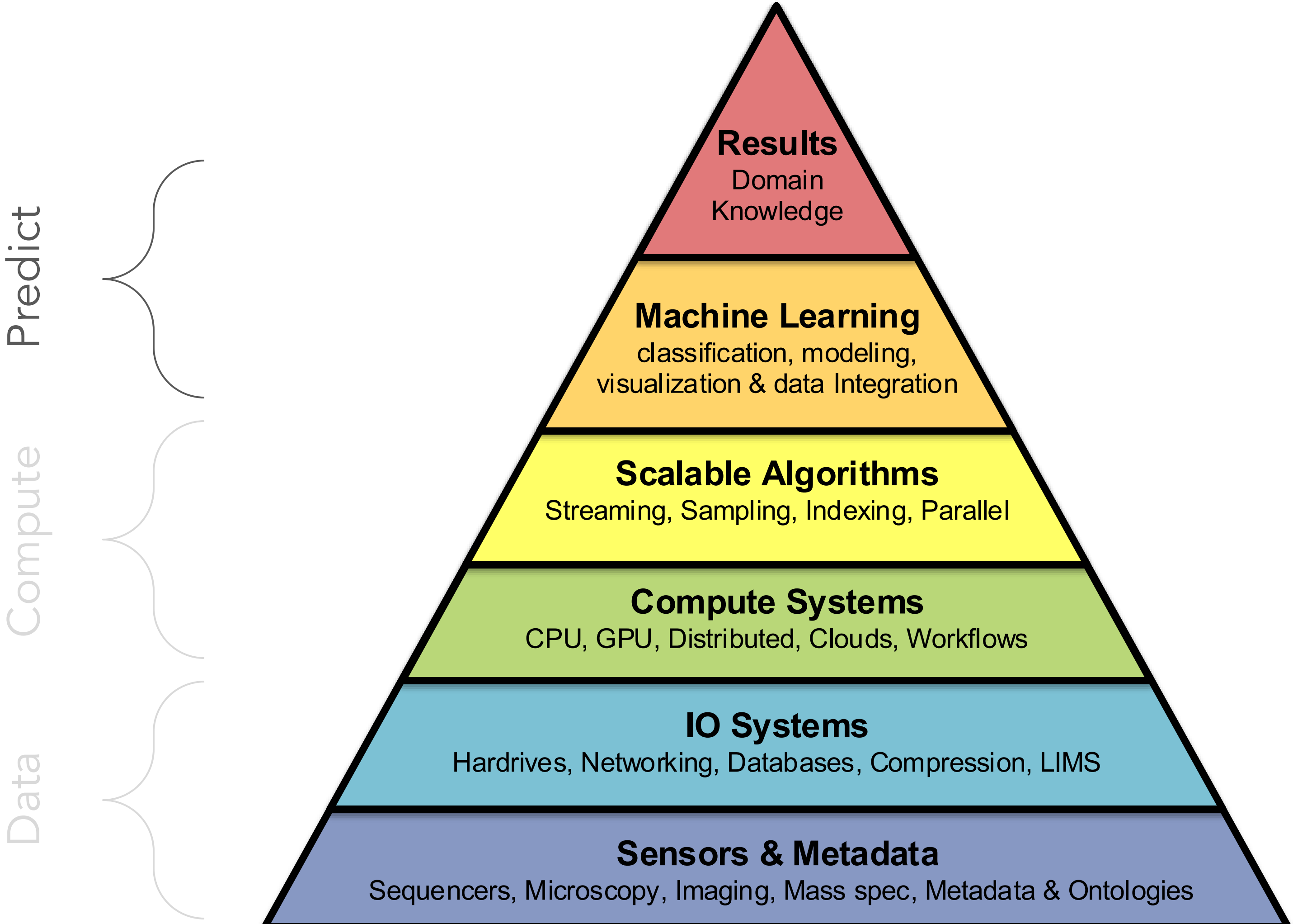
- 291 SV-disease associations identified
    - 148 (51%) absent from matched short-read callsets
    - 191 (66%) were lead variants compared with nearby SNVs
- 114 coding-region SV associations detected
    - 67 (59%) involved SVs absent from the short-read callset
    - 62 (54%) showed stronger association signals than nearby SNPs
    - Revealed previously unrecognized protein-altering effects

SVs are a major source of disease-relevant variation, contributing to the missing heritability in complex traits and diseases

Population-scale Long-read Sequencing in the All of Us Research Program

Kiran V Garimella, Qiuhui Li, Julie Wertz, Samuel K Lee, Fabio Cunial, Yongqing Huang, Yulia Mostovoy, Ryan Lorig-Roach, Adam English, Hang Su, Shawn Levy, Donna M Muzny, Chelsea Berngruber, Matt C Danzi, William T Harvey, Emily L LaPlante, Karynne Patterson, Allison N Rozanski, Sophie Schwartz, Beri Shifaw, Yuanyuan Wang, Isaac Wong, Isaac R. L. Xu, Shadi Zaheri, Stephan Zuchner, Xinchang Zheng, Shannon Dugan-Perez, Michal Izydorczyk, Heer Mehta, Richard A Gibbs, Lee Lichtenstein, Namrata Gupta, Niall Lennon, Stacey Gabriel, All of Us Research Program Long Read Working Group, Winston Timp, Kimberly F Doheny, Tara Dutka, Anjene Musick, Chia-Lin Wei, Fritz J Sedlazeck, Michael C Schatz, Michael E Talkowski, Evan E Eichler

doi: https://doi.org/10.1101/2025.10.02.25336942

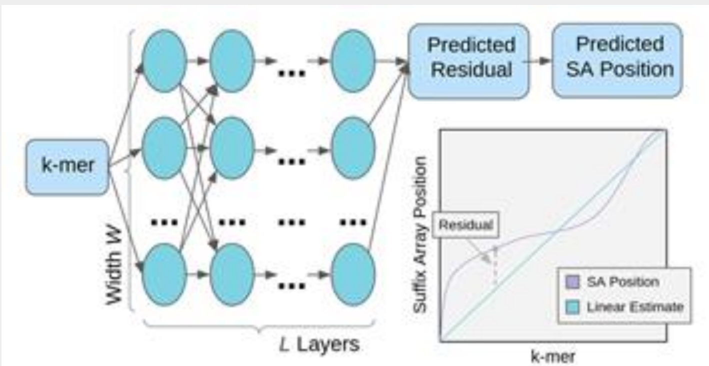


Biological data sciences in genome research

Schatz MC (2015) Genome Research doi: 10.1101/gr.191684.115

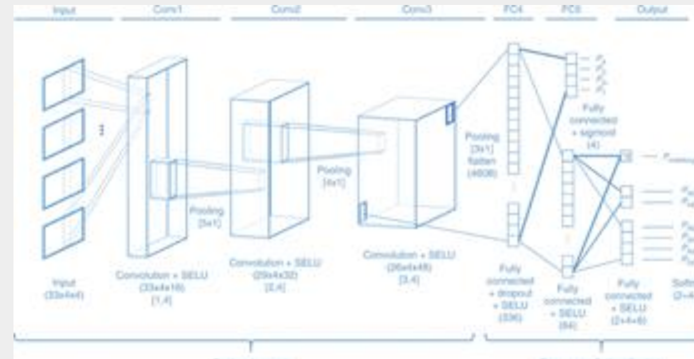
AI-powered Algorithms

ANN for Read Alignment



Sapling: Accelerating suffix array queries with learned data models
Kirsche et al. (2021) *Bioinformatics.*
doi: 10.1093/bioinformatics/btaa911

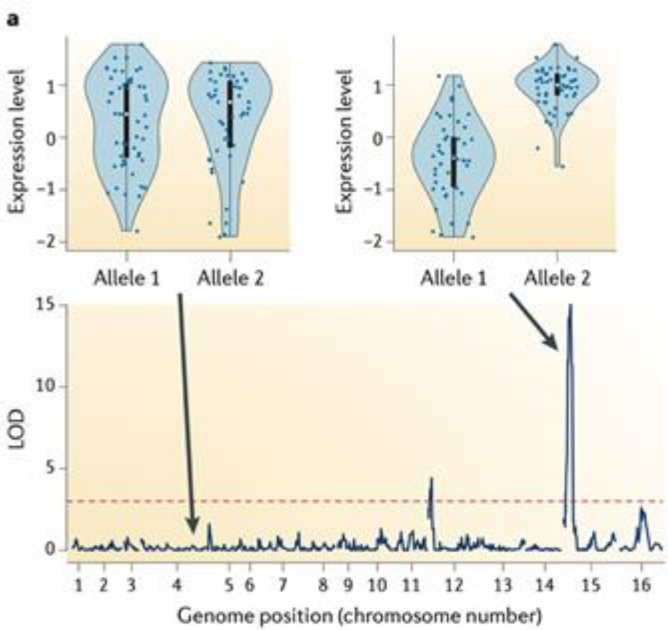
CNN for Variant Calling



A multi-task convolutional deep neural network for variant calling in single molecule sequencing
Luo et al. (2019) *Nature Communication.*
doi: 10.1038/s41467-019-09025-z

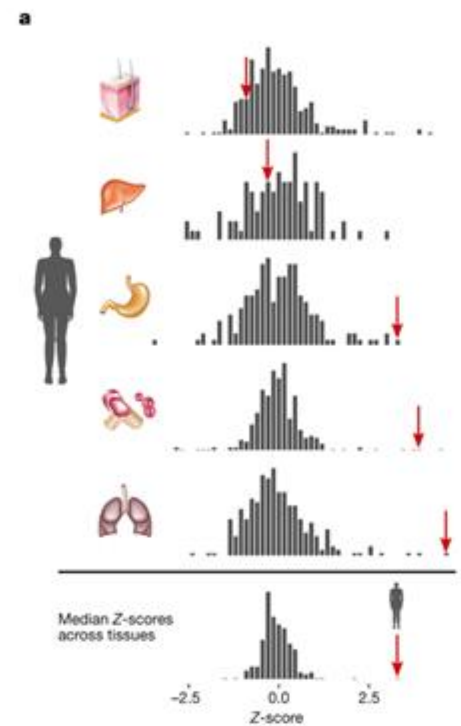
Genetics of complex traits and disease

Common Variants



The role of regulatory variation in complex traits and disease
Albert & Kruglyak (2015) Nat Rev Genet. doi: 10.1038/nrg3891

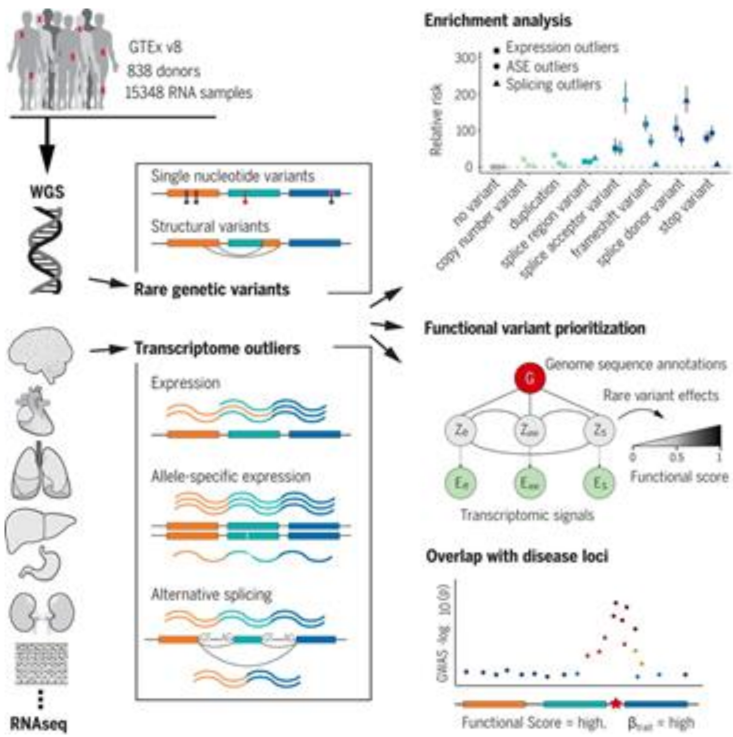
Rare Variants



The impact of rare variation on gene expression across tissues
Li *et al*. (2017) Nature. doi: 10.1038/nature24267

Rare variant analysis with Watershed

Alexis Battle



1. A set of variables **G**, representing the P observed genomic annotations aggregated over all RVs in the individual that are nearby the gene.
2. A set of binary latent variables $\mathbf{Z} = Z_1, \ldots, Z_K$ representing the unobserved functional regulatory status of the RVs on each of the K outlier signals.
3. A set of categorical nodes $\mathbf{E} = E_1, \ldots, E_K$ representing the observed outlier status of the gene for each of the K outlier signals.

- $Z|G \sim \mathrm{CRF}(\alpha, \beta_1, \ldots, \beta_k, \theta)$
- $E_k|Z_k \sim \mathrm{Categorical}(\phi_k)\ \forall\, k \in K$
- $\phi_k \sim \mathrm{Dirichlet}(C, \ldots, C)$
- $\beta_k \sim \mathrm{Normal}\left(0, \frac{1}{\lambda}\right)$

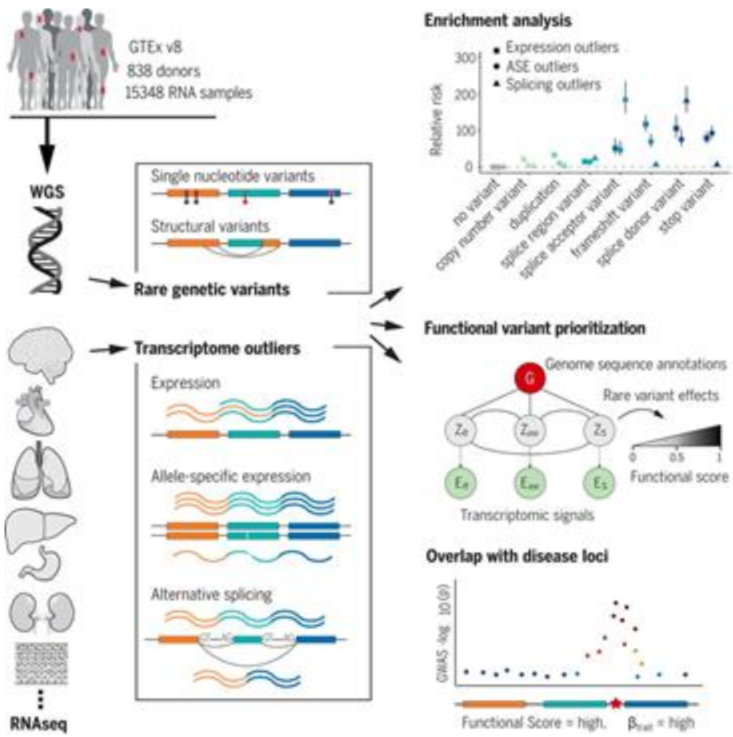
Transcriptomic signatures across human tissues identify functional rare genetic variation

Ferraro *et al* (2020) Science. doi: 10.1126/science.aaz5900

Rare variant analysis with Watershed-SV


Tanner Jensen


Bohan Ni

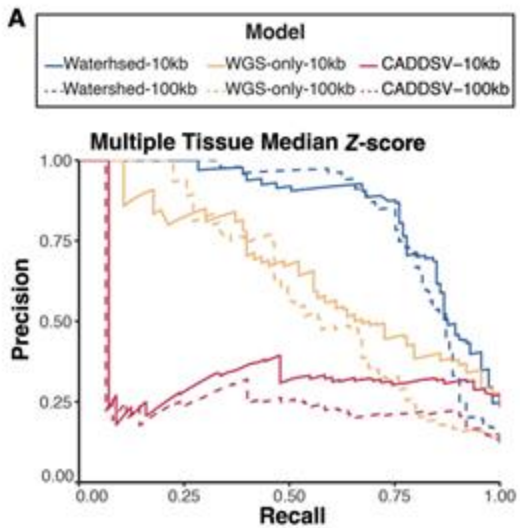


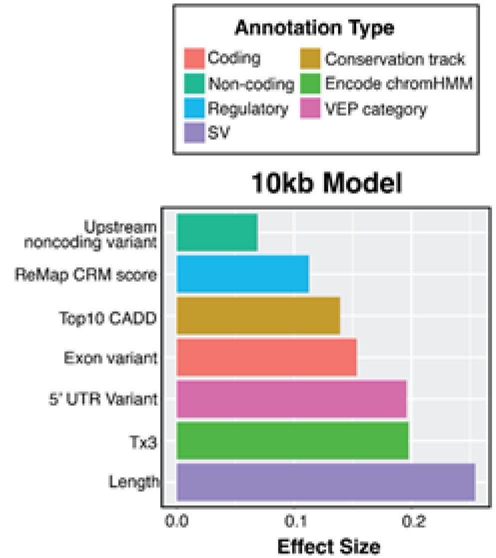
Genome Sequence Annotations	Metrics
SV characteristics	TYPE, length, VEP variant classifications, distance to gene, etc.
Summary statistics of Conservation and Functional Scores of regions covered	LINSIGHT score, CADD score, PhastCon score, etc.
Epigenomic Annotations and Regulatory Elements	Remap score 2020, non-cancer cell line Roadmap, etc.

Integration of transcriptomics and long-read genomics prioritizes structural variants in rare disease

Jensen*, Ni* *et al* (2025) *Genome Research.* doi: 10.1101/gr.279323.124

Watershed-SV on GTEx





- Train the model on rare variants and expression status in GTEx, focusing on rare variants near expression outliers
- By comparing held out "N2 pairs" of rare variants that occur in exactly two samples, evaluate the accuracy of Watershed-SV (blue) over a standard genomic annotation classification (without expression) (orange)
  - Watershed-SV is highly accurate identifying functionally relevant SVs
  - Key features include transcriptional status, variant length, PhastCon score, HeteroChromatin region

Integration of transcriptomics and long-read genomics prioritizes structural variants in rare disease
Jensen*, Ni* *et al* (2025) *Genome Research.* doi: 10.1101/gr.279323.124

Applying Watershed-SV model to UDN patients

Undiagnosed Diseases Network

Nanopore sequencing of **70 individuals from 51 rare disease families**
RNA-seq from blood and Cell-cultured fibroblast samples

Jensen*, Ni* *et al* (2025) *Genome Research*

Steven Montgomery

Euan Ashley

Tanner Jensen

Applying Watershed-SV model to UDN patients

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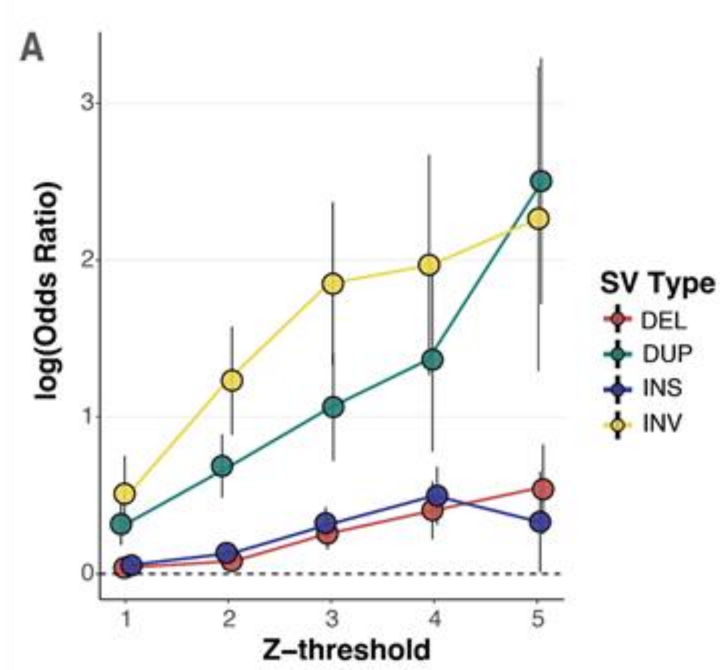


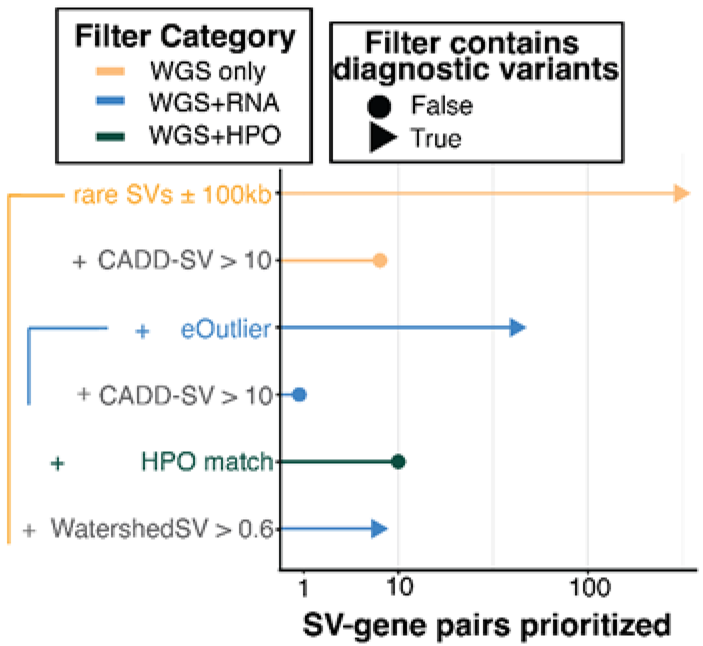
Steven Montgomery

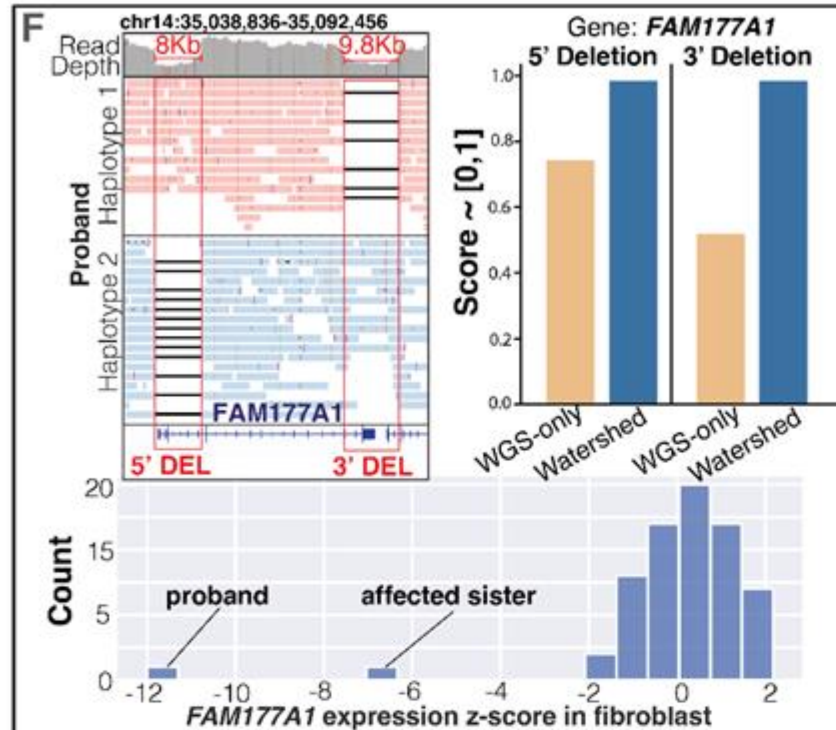
Euan Ashley

Tanner Jensen

Jensen*, Ni* *et al* (2025) *Genome Research*

Putative disease variant discovered in pair of siblings suffering from developmental delay, macrocephaly, and seizures





Jensen*, Ni* *et al* (2025) *Genome Research*

Next Steps


Method

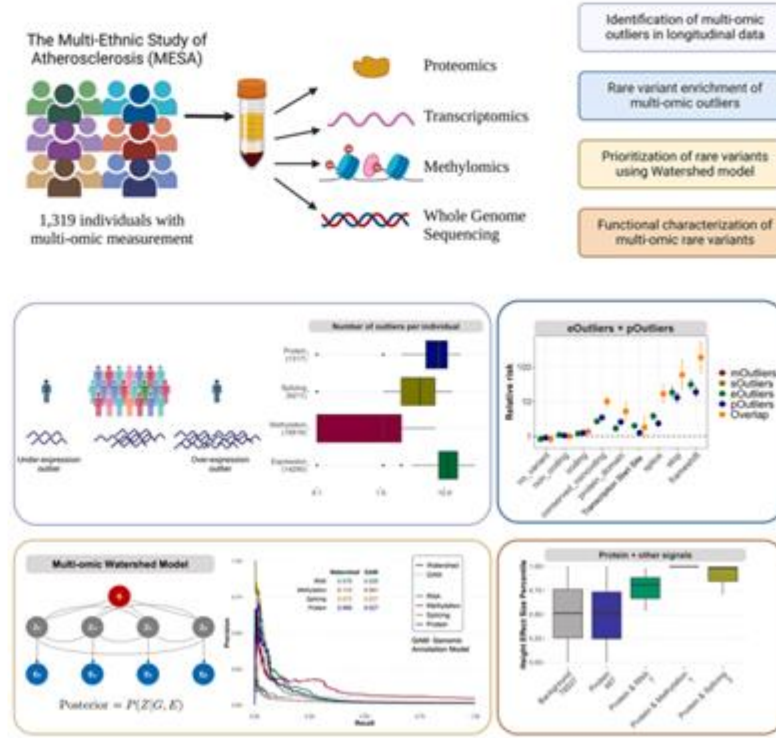
Integration of transcriptomics and long-read genomics prioritizes structural variants in rare disease

Tanner D. Jensen,[1,14] Bohan Ni,[2,14] Chloe M. Reuter,[3,4] John E. Gorzynski,[1,3,4] Sarah Fazal,[5] Devon Bonner,[3,4] Rachel A. Ungar,[1] Pagé C. Goddard,[1] Archana Raja,[3,4] Euan A. Ashley,[3,4] Jonathan A. Bernstein,[3,7] Stephan Zuchner,[5] Undiagnosed Diseases Network, Michael D. Greicius,[8] Stephen B. Montgomery,[1,9,10] Michael C. Schatz,[2] Matthew T. Wheeler,[3,4,11] and Alexis Battle[2,12,13]

Rare structural variants (SVs)—insertions, deletions, and complex rearrangements—can cause Mendelian disease, yet they remain difficult to accurately detect and interpret. We sequenced and analyzed Oxford Nanopore Technologies long-read genomes of 68 individuals from the undiagnosed disease network (UDN) with no previously identified diagnostic mutations from short-read sequencing. Using our optimized SV detection pipelines and 571 control long-read genomes, we detected 716 long-read rare (MAF < 0.01) SV alleles per genome on average, achieving a 2.4× increase from short reads. To characterize the functional effects of rare SVs, we assessed their relationship with gene expression from blood or fibroblasts from the same individuals and found that rare SVs overlapping enhancers were enriched (LOR = 0.46) near expression outliers. We also evaluated tandem repeat expansions (TREs) and found 14 rare TREs per genome; notably, these TREs were also enriched near overexpression outliers. To prioritize candidate functional SVs, we developed Watershed-SV, a probabilistic model that integrates expression data with SV-specific genomic annotations, which significantly outperforms baseline models that do not incorporate expression data. Watershed-SV identified a median of eight high-confidence functional SVs per UDN genome. Notably, this included compound heterozygous deletions in *FAM177A1* shared by two siblings, which were likely causal for a rare neurodevelopmental disorder. Our observations demonstrate the promise of integrating long-read sequencing with gene expression toward improving the prioritization of functional SVs and TREs in rare disease patients.

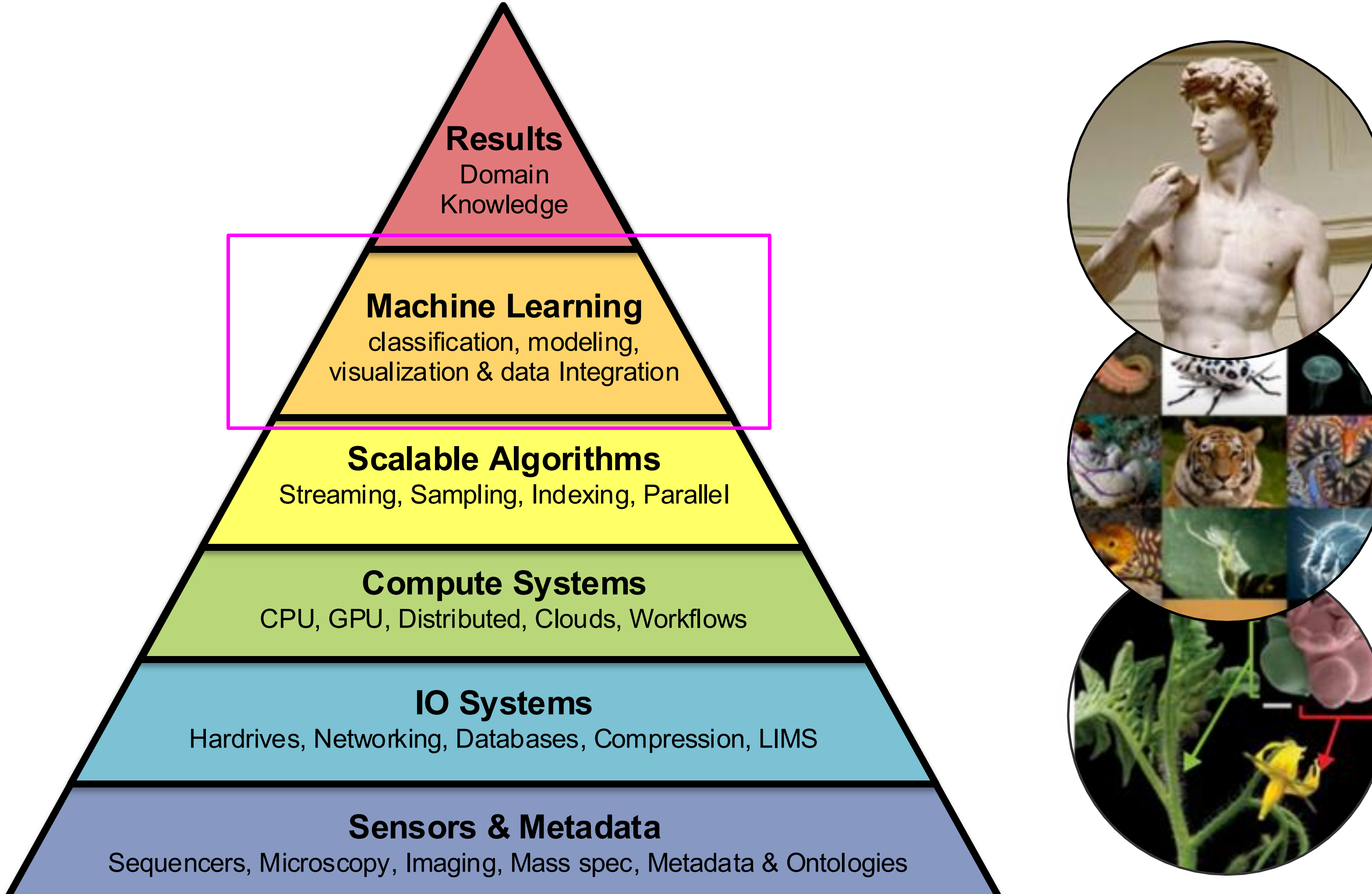
[Supplemental material is available for this article.]

Genome Research 1



Li et al. (2023) Cell Genomics. doi: 10.1016/j.xgen.2023.100401

- Watershed-SV on >1000 samples from MESA with matched DNA + RNA data
  - First investigation into the role of structural variations in atherosclerosis
  - Also applying Watershed-SV to other clinical samples



Biological data sciences in genome research

Schatz MC (2015) Genome Research doi: 10.1101/gr.191684.115

Genome Intelligence:
LLMs are breaking ground in genomics

DNABERT-2: Efficient Foundation Model and Benchmark For Multi-Species Genomes

Zhihan Zhou[†] **Yanrong Ji**[†] **Weijian Li**[†] **Pratik Dutta**[‡] **Ramana V Davuluri**[‡] **Han Liu**[†]
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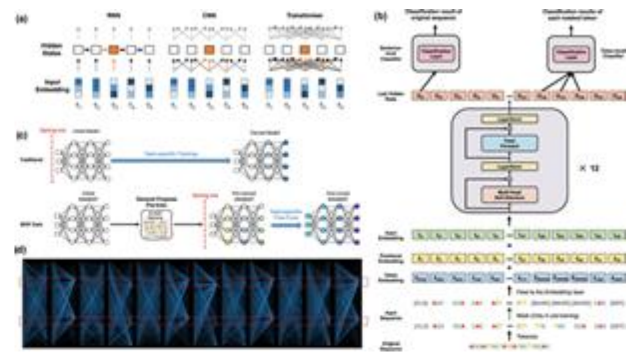
Abstract

Decoding the linguistic intricacies of the genome is a crucial problem in biology, and pre-trained foundational models such as DNABERT and Nucleotide Transformer have made significant strides in this area. Existing works have largely hinged on *k-mer*, fixed-length permutations of A, T, C, and G, as the *token* of the genome language due to its simplicity. However, we argue that the computation and sample inefficiencies introduced by k-mer tokenization are primary obstacles in developing large genome foundational models. We provide conceptual and empirical insights into genome tokenization, building on which we propose to replace k-mer tokenization with Byte Pair Encoding (BPE), a statistics-based data compression algorithm that constructs *tokens* by iteratively merging the most frequent co-occurring genome segment in the corpus. We demonstrate that BPE not only overcomes the limitations of k-mer tokenization but also benefits from the computational efficiency of non-overlapping tokenization. Based on these insights, we introduce DNABERT-2, a refined genome foundation model that adapts an efficient tokenizer and employs multiple strategies to overcome input length constraints, reduce time and memory expenditure, and enhance model capability. Furthermore, we identify the absence of a comprehensive and standardized benchmark for genome understanding as another significant impediment to fair comparative analysis. In response, we propose the Genome Understanding Evaluation, a comprehensive multi-species genome classification dataset that amalgamates 36 distinct datasets across 9 tasks, with input lengths ranging from 70 to 10000. Through comprehensive experiments on the GUE benchmark, we demonstrate that DNABERT-2 achieves comparable performance to the state-of-the-art model with 21× fewer parameters and approximately 92× less GPU time [1] in pre-training. Compared to DNABERT, while being 3× more efficient, DNABERT-2 outperforms it on 23 out of 28 datasets, with an average improvement of 6 absolute scores on GUE. The code, data, and pre-trained model are available at https://github.com/MAGICS-LAB/DNABERT_2.

1 Introduction

Transformer-based foundation models (Bommasani et al., 2022; Kenton & Toutanova, 2019; OpenAI, 2023) have witnessed significant progress in recent years, particularly exemplified by the advent of

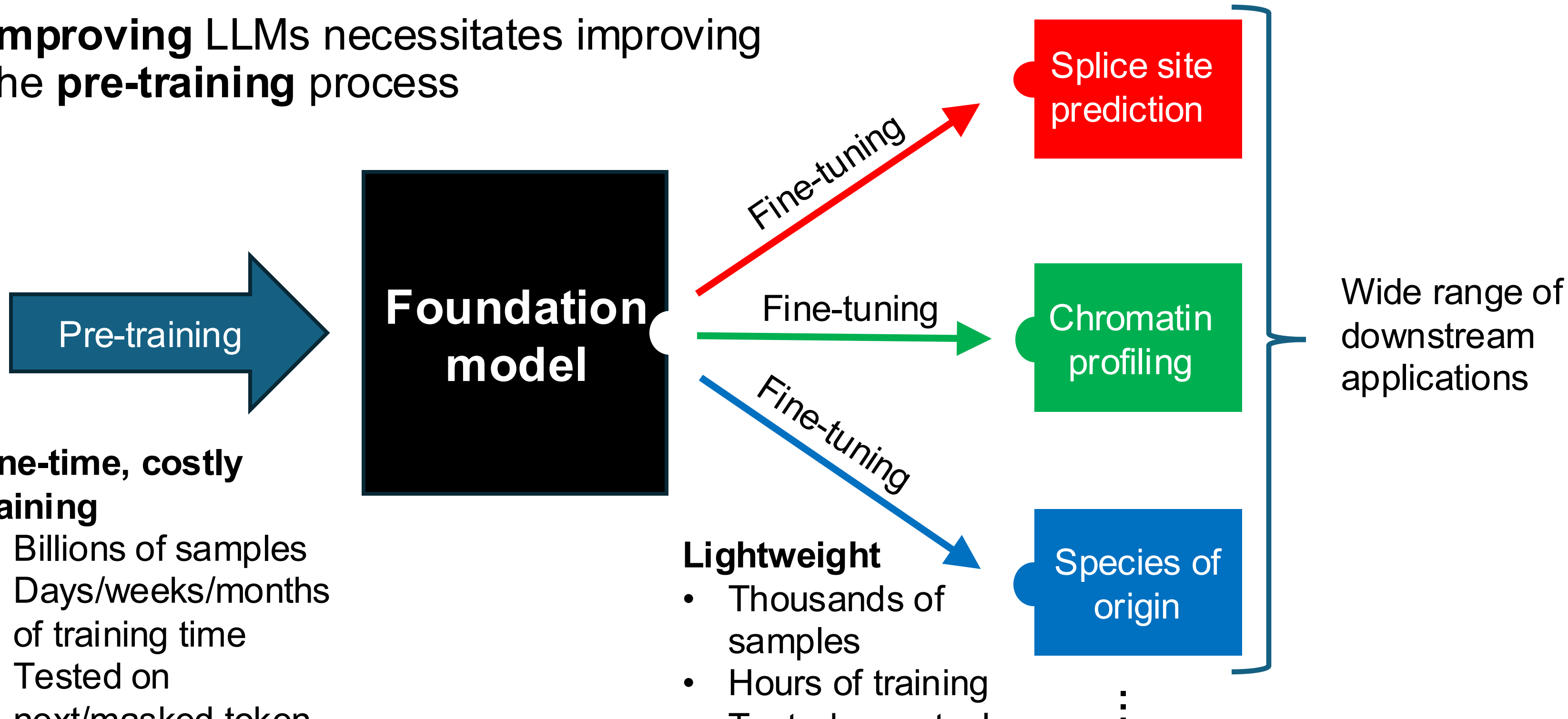


Species	Yeast	Mouse	Virus	Human			
Task	EMP	TF-M	CVC	TF-H	PD	CPD	SSP
DNABERT (3-mer)	49.54	57.73	62.23	64.43	84.63	**72.96**	84.14
DNABERT (4-mer)	48.59	59.58	59.87	64.41	82.99	71.10	84.05
DNABERT (5-mer)	48.62	54.85	63.64	50.46	84.04	72.03	84.02
DNABERT (6-mer)	49.10	56.43	55.50	64.17	81.70	71.81	84.07
NT-500M-human	45.35	45.24	57.13	50.82	85.51	66.54	79.71
NT-500M-1000g	47.68	49.31	52.06	58.92	86.58	69.13	80.97
NT-2500M-1000g	50.86	56.82	66.73	61.99	86.61	68.17	85.78
NT-2500M-multi	58.06	67.01	**73.04**	63.32	**88.14**	71.62	**89.36**
DNABERT-2	55.98	67.99	71.02	**70.10**	84.21	70.52	84.99
DNABERT-2♦	**58.83**	**71.21**	68.49	66.84	83.81	71.07	85.93

Table 4: The models' averaged performance on the 7 tasks in the GUE benchmark, including Epigenetic Marks Prediction (EMP), Transcription Factor Prediction on the Human genome and the Mouse genome (TF-H and TF-M), Covid Variants Classification (CVC), Promoter Detection (PD), Core Promoter Detection (CPD), and Splice Site Prediction (SSP).

Pre-training is key to the success of LLMs

Improving LLMs necessitates improving the **pre-training** process



One-time, costly training
- Billions of samples
- Days/weeks/months of training time
- Tested on next/masked token prediction

Lightweight
- Thousands of samples
- Hours of training
- Tested on actual task

Pre-training is key to the success of LLMs

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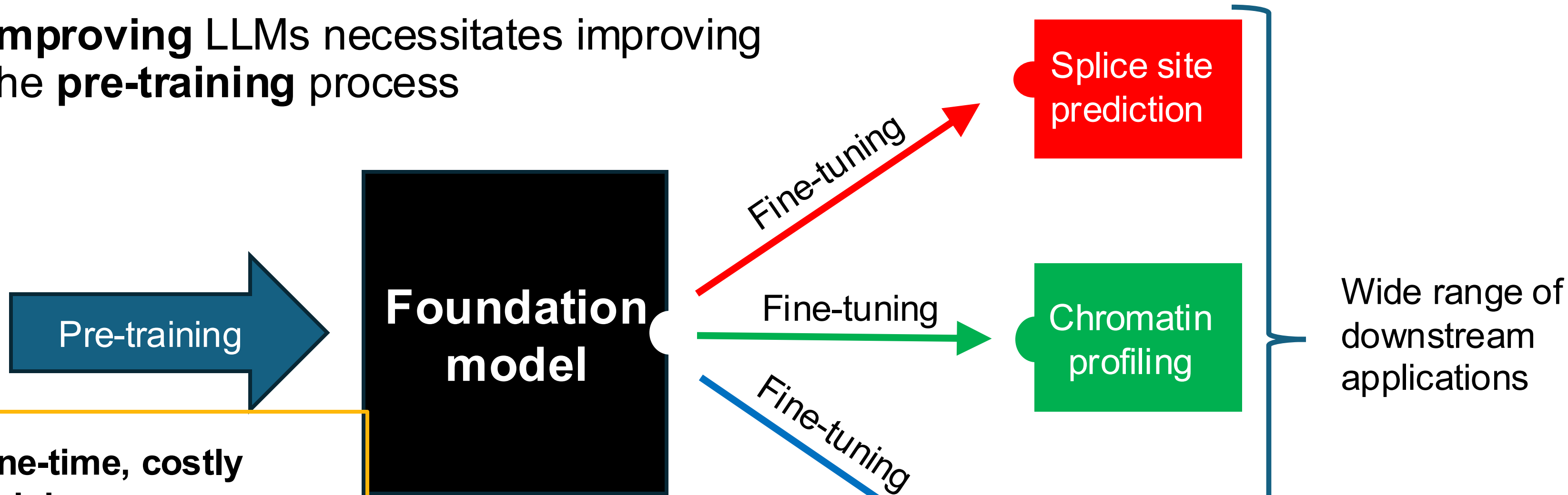

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- Hours of training
- Tested on actual task

JOHNS HOPKINS UNIVERSITY

Pre-Training Dataset Deduplication Improves Genomic LLMs

Mahler Revsine, Alex Ostrovsky, Xingyu Chen,
Daniel Khashabi, and Michael C. Schatz

Summary and future directions

AI is accelerating all aspects of genomics

- Data -> Compute -> Predict
- Impressive results predicting disease variants, molecular activities, health outcomes, and much more
- Even “classic” algorithms & statistical models benefit from learned index structures and other AI techniques

Healthy debate on use of specialized vs foundation models

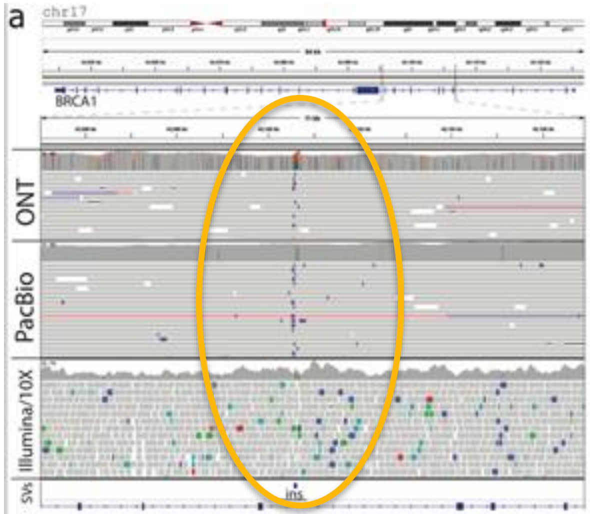
- Specialized models are often more efficient and interpretable, but can be trapped by their specialization

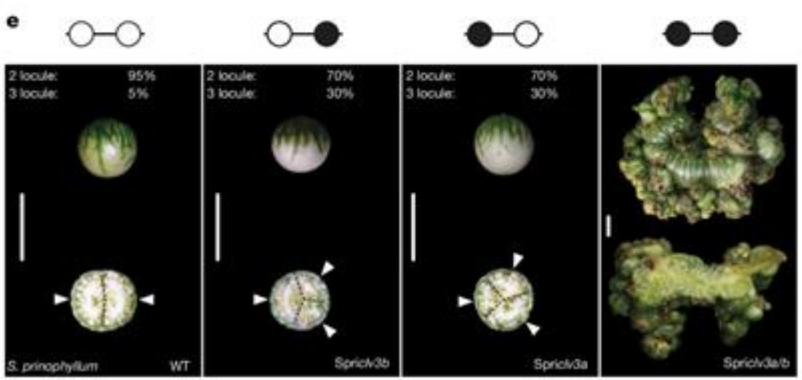
Academic computing can’t compete with Google/Meta/OpenAI with $$$, but we can be more clever

- Deduplication and other training optimizations make it more practical to scale to large collections of genomes and data

=> Expect widespread improvements in our ability to predict and understand many facets of biology

Where we need help





Interpreting rare and complex variations remains challenging

- AI methods are only as good as the training data, but we dont have extensive catalogs of complex variants with known outcomes

- “Standard” approaches (GWAS, QTL, etc) evaluate one variant at a time, but this is blind to epistatic interactions & network effects

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CSHL

Lippman Lab
McCombie Lab

Stanford

Montgomery Lab
Ashley Lab

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University of Toronto

Steven Gallinger

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PANCREATIC CANCER RESEARCH

NIH NATIONAL CANCER INSTITUTE

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Bloomberg Professors

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Thank you!

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